The `revtex4-1` document class of the American Physical Society *

Arthur Ogawa †

Version 4.1s, dated 2020/09/30

This file embodies the implementation of the APS REVTEX 4.1 document class for electronic submissions to journals.

The distribution point for this work is `http://publish.aps.org/revtex4/`, which contains fully unpacked, prebuilt runtime files and documentation.

Contents

1 Using REVTEX **3**
1.1 Bill of Materials . . . 3
1.1.1 Primary Source . . . 3
1.1.2 Generated by `latex ltxfront.dtx` . . . 3
1.1.3 Generated by `tex ltxfront.dtx` . . . 3
1.1.4 Documentation . . . 3
1.1.5 Auxiliary . . . 3

2 Code common to all modules **3**

3 The doc module `doc` **4**
3.1 The Preamble . . . 4
3.1.1 Docstrip and info directives . . . 5
3.2 The "Read Me" File . . . 5
3.3 A Gnuly Created Sample Document . . . 8
3.4 The Document Body . . . 9

4 Using this package **9**
4.1 Invoking the package . . . 9

5 Frontmatter commands **10**

*Work under hire to The American Physical Society.

†First revision of REVTEX4.0 (unreleased) by David Carlisle

6 The Front Matter **10**
6.1 Setting the title . . . 10
6.2 Specifying a date . . . 10
6.3 Specifying authors and affiliations . . . 10
6.3.1 Collaborations . . . 12
6.3.2 Footnotes for authors, collaborations, affiliations or title . . . 12
6.3.3 Specifying first names and surnames . . . 14
6.4 The abstract . . . 14
6.5 PACS codes . . . 14
6.6 Keywords . . . 14
6.7 Institutional report numbers . . . 15
6.8 maketitle . . . 15

7 Compatability with LaTeX's Required Packages **15**

8 Implementation of package **15**
8.1 Beginning of the `package` DOCSTRIP module . . . 15
8.2 Prerequisites . . . 16
8.3 Options . . . 16
8.4 Process package options . . . 17

9 Kernel of Front Matter **18**
9.1 Initialization . . . 18
9.2 The title command . . . 18
9.3 The author, affiliation, and related commands . . . 19
9.3.1 Commands for author or collaboration . . . 19
9.3.2 Commands for affiliation . . . 22
9.3.3 Commands for auxiliary information . . . 29
9.3.4 Procedures for author, collaboration, and affiliation . . . 30
9.4 The keywords command . . . 34
9.5 The `\date` command and related commands . . . 34
9.6 The pacs command . . . 34
9.7 The `\preprint` command . . . 35
9.8 The `abstract` environment . . . 35
9.9 Formatting the title block . . . 38
9.9.1 Authors and affiliations in superscriptaddress style . . . 41
9.9.2 Authors and affiliations in groupedaddress style . . . 47

10 Package defaults **51**
10.1 Printing out the "list-of" elements . . . 54
10.2 Syntax switch . . . 55
10.3 End of the `ltxfront` kernel . . . 56
10.4 Remainder of the `ltxfront` package . . . 56

Index **58**

1 Using REVTEX

The file `README` has retrieval and installation information.

User documentation is presented separately in `auguide.tex`.

The file `template.aps` is a boilerplate file.

1.1 Bill of Materials

Following is a list of the files in this distribution arranged according to provenance.

1.1.1 Primary Source

One single file generates all.

`ltxfront.dtx`

1.1.2 Generated by `latex ltxfront.dtx`

Typesetting the source file under LaTeX generates the readme and the documentation.

`00readme`

1.1.3 Generated by `tex ltxfront.dtx`

Typesetting the installer generates the package files.

`ltxfront.sty`

1.1.4 Documentation

The following are the online documentation:

`ltxfront.pdf`

1.1.5 Auxiliary

The following are auxiliary files generated in the course of running LaTeX:

`ltxfront.aux ltxfront.idx ltxfront.ind ltxfront.log ltxfront.toc`

2 Code common to all modules

We want to require only one place in this file where the version number is stated, and we also want to ensure that the version number is embedded into every generated file.

Now we declare that these files can only be used with LaTeX 2_ε. An appropriate message is displayed if a different TeX format is used.

```
%<*doc|package>
\NeedsTeXFormat{LaTeX2e}[1995/12/01]%
%</doc|package>
```

As desired, the following modules all take common version information:

```
%<kernel&!package&!doc>\typeout{%
%<*package|doc>
\ProvidesFile{%
%</package|doc>
%<*kernel|package|doc>
ltxfront%
%</kernel|package|doc>
%<*doc>
.dtx%
%</doc>
%<package>.sty%
%<*package|doc>
}%
%</package|doc>
```

The following line contains, for once and for all, the version and date information. By various means, this information is reproduced consistently in all generated files and in the typeset documentation.

```
%<*doc|package|kernel>
%<version>
 [2020/09/30 4.1s frontmatter package (AO,DPC)]% \fileversion
%</doc|package|kernel>
%<kernel&!package&!doc>}%
```

3 The doc module `doc`

This module, consisting of the present section, typesets the programmer's documentation, generating the `00readme` and sample document as needed.

Because the only uncommented-out lines of code at the beginning of this file constitute the `doc` module itself, we can simply typeset the `.dtx` file directly, and there is thus rarely any need to generate the "doc" DOCSTRIP module. Module delimiters are nonetheless required so that this code does not find its way into the other modules.

The `\end{document}` command concludes the typesetting run.

```
%<*doc>
```

3.1 The Preamble

The programmers documentation is formatted with the `ltxdoc` class with local customizations, and with the usual code line indexing.

```
\documentclass{ltxdoc}
\RequirePackage{ltxdocext}%
\let\url\undefined
```

```
\RequirePackage[colorlinks=true,linkcolor=blue]{hyperref}%
\makeatletter
\@ifundefined{package@font}{}%
  {\expandafter\RequirePackage\expandafter{\csname package@font\endcsname}}
\makeatother
\CodelineIndex\EnableCrossrefs % makeindex -s gind.ist ltxfront
\RecordChanges % makeindex -s gglo.ist -o ltxfront.gls ltxfront.glo
```

3.1.1 Docstrip and info directives

We use so many DOCSTRIP modules that we set the `StandardModuleDepth` counter to 1.

```
\setcounter{StandardModuleDepth}{1}
```

The following command retrieves the date and version information from this file.

```
\expandafter\GetFileInfo\expandafter{\jobname.dtx}%
```

3.2 The "Read Me" File

As promised above, here is the contents of the "Read Me" file. That file serves a double purpose, since it also constitutes the beginining of the programmer's documentation. What better thing, after all, to have appear at the beginning of the typeset documentation?

A good discussion of how to write a ReadMe file can be found in Engst, Tonya, "Writing a ReadMe File? Read This" *MacTech* October 1998, p. 58.

Note the appearance of the `\StopEventually` command, which marks the dividing line between the user documentation and the programmer documentation.

The usual user will not be asked to do a full build, not to speak of the bootstrap. Instructions for carrying these processes begin the programmer's manual.

```
\begin{filecontents*}{00readme.tex}
\title{%
 A \LaTeX\ Package for\\the frontmatter and title page%
 \thanks{%
  This file has version number \fileversion,
  last revised \filedate.%
 }%
 \thanks{%
  Version \fileversion\ \copyright\ 2009 The American Physical Society
 }%
}%

\author{%
 Arthur Ogawa%
 \thanks{\texttt{mailto:arthur\_ogawa at sbcglobal.net}}%
}%
%\iffalse
% For version number and date,
% search on "\fileversion" in the .dtx file,
```

```
% or see the end of the 00readme file.
%\fi
\maketitle

This file embodies the \classname{ltxfront} package,
the implementation and its user documentation.

The distribution point for this work is
\url{http://publish.aps.org/revtex},
which contains fully unpacked, prebuilt runtime files and documentation.

The \classname{ltxfront} package was commissioned by the American Physical Society
and is distributed under the terms of the \LaTeX\ Project Public License,
the same license under which all the portions of \LaTeX\ itself is distributed.
Please see \url{http://ctan.tug.org/macros/latex/base/lppl.txt} for details.

To use this document class, you must have a working
\TeX\ installation equipped with \LaTeXe\
and possibly pdftex and Adobe Acrobat Reader or equivalent.

To install, retrieve the distribution,
unpack it into a directory on the target computer,
and move the file \file{ltxfront.sty}
into a location in your filesystem where it will be found by \LaTeX.

To use, read the user documentation \file{ltxfront.pdf}.

\tableofcontents

\section{Processing Instructions}

The package file \file{ltxfront.sty}
is generated from this file, \file{ltxfront.dtx},
using the {\sc docstrip} facility of \LaTeX
via |tex ltxfront.dtx|.
The typeset documentation that you are now reading is generated from
the same file by typesetting it with \LaTeX\ or pdftex
via |latex ltxfront.dtx| or |pdflatex ltxfront.dtx|.

\subsection{Build Instructions}

You may bootstrap this suite of files solely from \file{ltxfront.dtx}.
Prepare by installing \LaTeXe\ (and either tex or pdftex) on your computer,
then carry out the following steps:
\begin{enumerate}

\item
Within an otherwise empty directory,
typeset \file{ltxfront.dtx} with \TeX\ or pdftex,
thereby generating the package file \file{ltxfront.sty}.
```

```
105
106 \item
107 Next typeset \file{ltxfront.dtx} with \LaTeX\ or pdflatex;
108 you will obtain the typeset documentation you are now reading,
109 along with
110 the sample document \file{ltxfront-test.tex}
111 and the file \file{00readme}.
112
113 Note: you will have to run \LaTeX, then
114 \file{makeindex} \texttt{-s gind.ist ltxfront.idx}, then
115 \file{makeindex} \texttt{-s gglo.ist -o ltxfront.gls ltxfront.glo}, then
116 \LaTeX\ again in order to obtain a valid index and table of contents.
117 \item
118 Now typeset \file{ltxfront.dtx} with \TeX (not \LaTeX),
119 thereby generating the package file \file{ltxfront.sty}.
120 \item
121 Install the following files into indicated locations within your
122 TDS-compliant \texttt{texmf} tree (you may need root access):
123 \begin{itemize}
124 \item
125 \file{$TEXMF/}\file{tex/}\file{latex/}\file{revtex/}\classname{ltxfront.sty}
126 \item
127 \file{$TEXMF/}\file{source/}\file{latex/}\file{revtex/}\classname{ltxfront.dtx}
128 \item
129 \file{$TEXMF/}\file{doc/}\file{latex/}\file{revtex/}\classname{ltxfront.pdf}
130 \end{itemize}
131 where \file{$TEXMF/} stands for \file{texmf-local/}, or some other \texttt{texmf} tree
132 in your installation.
133 \item
134 Run \texttt{mktexlsr} on \file{$TEXMF/} (you may need root access).
135 \item
136 Build and installation are now complete;
137 now put a \cmd\usepackage\texttt{\{ltxfront\}} in your document preamble!
138 \end{enumerate}
139
140 \subsection{Change Log}
141 \changes{4.0b}{1999/06/20}{Change handling of \cs{@author} to prevent parsing problems. (Helps
142 \changes{4.0b}{1999/06/20}{Gathered all code for the abstract environment together and abstract
143 \changes{4.0b}{1999/06/20}{Separate option now controls production}
144 \changes{4.0b}{1999/06/20}{Set \cs{footnote} to \LaTeX\ standard version for this scope.}
145 \changes{4.0c}{1999/11/13}{grid changes with ltxgrid}
146 \changes{4.0d}{1998/01/31}{\cs{comma@space}}
147 \changes{4.0d}{1998/01/31}{change name from \cs{tempaffiliation}}
148 \changes{4.0d}{1998/01/31}{hookify}
149 \changes{4.0d}{1998/01/31}{If empty \cs{abstractname} omit spacing}
150 \changes{4.0d}{1998/01/31}{PACS boxed to width of abstract}
151 \changes{4.0e}{2000/09/20}{Keywords on same footing as PACS}
152 \changes{4.0e}{2000/09/20}{Warn if no production}
153 \changes{4.0f}{2001/02/13}{Actually append \cs{AU@grp} to \cs{@AAClist}.}
154 \changes{4.0g}{1998/06/10}{multiple preprint commands}
```

```
\changes{4.0h}{1998/06/25}{Changes to support groups of affiliations}
\changes{4.0h}{1998/06/25}{new hook}
\changes{4.0l}{1998/09/01}{add optional argument handling back}
\changes{4.0l}{1998/09/01}{Changes to add collaboration processing, which now means doing comma
\changes{4.0l}{1998/09/01}{macro added}
\changes{4.1a}{2008-06-30}{(AO, 443) create an entry in the PDF bookmarks for the abstract via
\changes{4.1a}{2008-07-01}{(AO, 496) remove default prependation to \cs{email}.}
\changes{4.1a}{2008-07-01}{(AO, 496) remove default prependation to \cs{homepage}.}
\changes{4.1a}{2008/06/16}{removed code that had been commented out}
\changes{4.1b}{2008/06/29}{(AO, 455) provide a \texttt{description} environment for the abstrac
\changes{4.1b}{2008/08/02}{(AO, 486) \cs{collaboration} to work with \texttt{groupedaddress}}
\changes{4.1b}{2008/08/02}{Change csname \cs{AFS@opr} to \cs{AFG@opr}}
\changes{4.1b}{2008/08/02}{Make the syntax of this package switch optional}%
\changes{4.1b}{2008/08/02}{Provide for diagnostics via \cs{frontmatterverbose@sw}}
\changes{4.1b}{2008/08/02}{Use \cs{@affil@def} for all access to \cs{@affil}}
\changes{4.1b}{2008/08/02}{Use \cs{frontmatter@init} as the vehicle for all initialization}
\changes{4.1b}{2008/08/04}{(AO, 443) PDF Bookmark for title page}%
\changes{4.1b}{2008/08/04}{Provide facility \cs{affil@cutoff}: if there is only one affiliation
\changes{4.1c}{2008/08/04}{Hyperref compatibility: take \cs{maketitle} back.}
\changes{4.1c}{2008/08/04}{Parametrize the production after the address}%
\changes{4.1c}{2008/08/04}{Produce PACS, keywords only if specified}%
\changes{4.1d}{2008/10/17}{(AO, 410) compare \cs{@tempa} to \cs{blankaffiliation}.}%
\changes{4.1f}{2009/07/07}{(AO, 515) Use \cs{set@footnotefont}, which is defined in ltxutil}
\changes{4.1g}{2009/10/05}{(AO, 535) Robustify \cs{footnote} while \cs{class@warn} is expanding
\changes{4.1g}{2009/10/05}{(AO, 539) Robustify \cs{class@warn} against TOC processing}%
\changes{4.1i}{2009/10/18}{(AO, 540) More runtime diagnostics via \cs{frontmatterverbose@sw}}
\changes{4.1i}{2009/10/23}{(AO, 540) Rationalize code that appends to \cs{AF@grp}}
\changes{4.1i}{2009/10/23}{(AO, 540) Optimize boolean logic }
\changes{4.1i}{2009/10/23}{(AO, 544) Class options runinaddress and unsortedaddress stopped wor
\changes{4.1m}{2009/11/20}{(AO, 563) \cs{collaboration} to work with both \texttt{groupedaddres
\changes{4.1m}{2009/11/20}{(AO, 563) More complete runtime diagnostics via \cs{frontmatterverbo
\changes{4.1m}{2009/11/20}{(AO, 563) Let \cs{@collaboration@gobble} parse all three arguments o
\changes{4.1n}{2009/11/30}{(AO, 567) Handle \cs{noaffiliation} case}
\changes{4.1n}{2009/11/30}{(AO, 567) Handle punctuation for \classoption{superscriptaddress} st
\changes{4.1n}{2009/12/03}{(AO) remove http:// from \cs{href} call}
\changes{4.1n}{2010/01/02}{(AO, 571) Interface \cs{set@footnotewidth} for determining the set w
\changes{4.1n}{2010/01/02}{(AO, 571) allow split after last line of footnote}%
\changes{4.1n}{2010/01/02}{(AO, 572) Independent footnote counter for title block. Abstract foo

\end{filecontents*}
```

3.3 A Gnuly Created Sample Document

```
\begin{filecontents*}{ltxfront-test.tex}
\documentclass{article}% use any combination of titlepage, twocolumn
\usepackage[]{ltxfront}% use inactive to turn off features
\begin{document}

\title{Gnu Veldt Cuisine}

```

```
\author{G. Picking}
\thanks{Supported by a grant from the GSF.}
\affiliation{Acme Kitchen Products}

\author{R. Dillo}
\thanks{On leave during 1985.}
\affiliation{Cordon Puce School}

\date{24 July 1984}%
\revised{5 January 1985}%

\maketitle

\section{Introduction}%
We reproduce most features of \LaTeX\ manual, Figure C.4, p. 183.
\end{document}
\end{filecontents*}
```

3.4 The Document Body

Here is the document body, containing only a `\DocInput` directive—referring to this very file. This very cute self-reference is a common `ltxdoc` idiom.

```
\begin{document}%
\expandafter\DocInput\expandafter{\jobname.dtx}%
% ^^A\PrintChanges
\end{document}

%</doc>
```

4 Using this package

Once this package is installed on your filesystem, you can employ it in adding functionality to LaTeX by invoking it in your document or document class.

4.1 Invoking the package

In your document, you can simply call it up in your preamble:

```
%\documentclass{book}%
%\usepackage{ltxfront}%
%\begin{document}
%⟨your document here⟩
%\end{document}
```

However, the preferred way is to invoke this package from within your customized document class:

```
%\NeedsTeXFormat{LaTeX2e}[1995/12/01]%
%\ProvidesClass{myclass}%
%\RequirePackage{ltxfront}%
```

```
%\LoadClass{book}%
%⟨class customization commands⟩
%\endinput
```

Once loaded, the package gives you acccess to certain procedures, usually to be invoked by a LaTeX command or environment.

5 Frontmatter commands

A document using this package will use commands like `\title`, `\author`, `\thanks`, the `abstract` environment and `\maketitle`. These are part of LaTeX's standard document classes. But this package supplements LaTeX syntax with additional commands, such as `\affiliation`, `\collaboration`, `\email`, `\homepage`, and others.

Furthermore, you can have multiple `\author`, `\affiliation`, and `\collaboration` commands, each with their own set of `\email`, `\homepage`, etc. commands.

6 The Front Matter

This section describes the macros this package provides for formatting the front matter of an article. The behavior and usage of these macros can be quite different from those provided in LaTeX 2_ε.

6.1 Setting the title

The title of the manuscript is simply specified by using the `\title{`⟨*title*⟩`}` macro. A `\\` may be used to put a line break in a long title. The `\title` commnd takes an optional argument, which you will use to give an abbreviated title for use in the running header.

6.2 Specifying a date

The `\date`⟨*date*⟩ command outputs the date on the manuscript. Using `\today` will cause LaTeX to insert the current date whenever the file is run:

```
\date{\today}
```

6.3 Specifying authors and affiliations

Commands specifying author and affilliation have been improved to save labor for authors and in production. Authors and affiliations are arranged into groupings called, appropriately enough, *author groups*. Each author group is a set of authors who share the same set of affiliations. Author names are specified with the `\author` macro while affiliations (or addresses) are specified with the `\affiliation` macro. Author groups are specified by sequences of `\author` macros followed by `\affiliation` macros. An `\affiliation` macro applies to

all previously specified `\author` macros which don't already have an affiliation supplied.

For example, if Bugs Bunny and Roger Rabbit are both at Looney Tune Studios, while Mickey Mouse is at Disney World, the markup would be:

```
% \author{Bugs Bunny}
% \author{Roger Rabbit}
% \affiliation{Looney Tune Studios}
% \author{Mickey Mouse}
% \affiliation{Disney World}
```

The default is to display this as

Bugs Bunny and Roger Rabbit
Looney Tune Studios
Mickey Mouse
Disney World

This layout style for displaying authors and their affiliations is chosen by selecting the class option `groupedaddress`. This option is the default for all APS journal styles, so it does not need to be specified explicitly. The other major way of displaying this information is to use superscripts on the authors and affiliations. This can be accomplished by selecting the class option `superscriptaddress`. To achieve the display

Bugs Bunny,[1] Roger Rabbit,[1,2] and Mickey Mouse[2]
[1]*Looney Tune Studios*
[2]*Disney World*

one would use the markup

```
% \author{Bugs Bunny}
% \affiliation{Looney Tune Studios}
% \author{Roger Rabbit}
% \affiliation{Looney Tune Studios}
% \affiliation{Disney World}
% \author{Mickey Mouse}
% \affiliation{Disney World}
```

Note that this package takes care of any commas and *and*'s that join the author names together and font selection, as well as any superscript numbering. Only the author names and affiliations should be given within their respective macros.

There is a third class option, `unsortedaddress`, for controlling author/affiliation display. The default `groupedaddress` will actually sort authors into the appropriate author groups if one chooses to specify an affiliation for each author. The markup:

```
% \author{Bugs Bunny}
% \affiliation{Looney Tune Studios}
% \author{Mickey Mouse}
```

```
% \affiliation{Disney World}
% \author{Roger Rabbit}
% \affiliation{Looney Tune Studios}
```

will result in the same display as for the first case given above even though Roger Rabbit is specified after Mickey Mouse. To avoid Roger Rabbit being moved into the same author group as Bugs Bunny, use the `unsortedaddress` option instead. In general, it is safest to list authors in the order they should appear and specify affiliations for multiple authors rather than one at a time. This will afford the most independence for choosing the display option. Finally, it should be mentioned that the affiliations for the `superscriptaddress` are presented and numbered in the order that they are encountered. This means that the order will usually follow the order of the authors. An alternative ordering can be forced by including a list of `\affiliation` commands before the first `\author` in the desired order. Then use the exact same text for each affilation when specifying them for each author.

If an author doesn't have an affiliation, the `\noaffiliation` macro may be used in the place of an `\affiliation` macro.

6.3.1 Collaborations

A collaboration name can be specified with the `\collaboration` command. This is very similar to the `\author` command. It should appear at the end of an author group. The collaboration name will appear centered in parentheses. You may have multiple author groups, each with its own `\collaboration`. Because collaborations don't normally have affiliations, one needs to follow the `\collaboration` with `\noaffiliation`.

6.3.2 Footnotes for authors, collaborations, affiliations or title

Often one wants to specify additional information associated with an author, collaboration, or affiliation such an e-mail address, an alternate affiliation, or some other ancillary information. This package introduces several new macros just for this purpose. They are:

- `\email[`⟨*optional text*⟩`]{`⟨*e-mail address*⟩`}`
- `\homepage[`⟨*optional text*⟩`]{`⟨*URL*⟩`}`
- `\altaffiliation[`⟨*optional text*⟩`]{`⟨*affiliation*⟩`}`
- `\thanks{`⟨*miscellaneous text*⟩`}`

In the first three, the *optional text* will be prepended before the actual information specified in the required argument. `\email` and `\homepage` each have a default text for their optional arguments ('Electronic address:' and 'URL:' respectively). The `\thanks` macro should only be used if one of the other three do not apply. Any author name can have multiple instances of these four commands. Note that unlike the `\affiliation` macro, these macros only apply to the `\author` that

directly precedes it. Any `\affiliation` *must* follow the other author-specific macros. A typical usage might be as follows:

```
% \author{Bugs Bunny}
% \email[E-mail me at: ]{bugs@looney.com}
% \homepage[Visit: ]{http://looney.com/}
% \altaffiliation[Permanent address: ]
%                       {Warner Brothers}
% \affiliation{Looney Tunes}
```

This would result in the footnote "E-mail me at: `bugs@looney.com`, Visit: `http://looney.com/`, Permanent address: Warner Brothers" being attached to Bugs Bunny. Note that:

- Only an e-mail address, URL, or affiliation should go in the required argument in the curly braces.
- The font is automatically taken care of.
- An explicit space is needed at the end of the optional text if one is desired in the output.
- Use the optional arguments to provide customized text only if there is a good reason to.

The `\collaboration` , `\affiliation`, or even `\title` can also have footnotes attached via these commands. If any ancillary data (`\thanks`, `\email`, `\homepage`, or `\altaffiliation`) are given in the wrong context (e.g., before any `\title`, `\author`, `\collaboration`, or `\affiliation` command has been given), then a warning is given in the TEX log, and the command is ignored.

Duplicate sets of ancillary data are merged, giving rise to a single shared footnote. However, this only applies if the ancillary data are identical: even the order of the commands specifying the data must be identical. Thus, for example, two authors can share a single footnote indicating a group e-mail address.

Duplicate `\affiliation` commands may be given in the course of the front matter, without the danger of producing extraneous affiliations on the title page. However, ancillary data should be specified for only the first instance of any particular institution's `\affiliation` command; a later instance with different ancillary data will result in a warning in the TEX log.

It is preferable to arrange authors into sets. Within each set all the authors share the same group of affiliations. For each author, give the `\author` (and appropriate ancillary data), then follow this author group with the needed group of `\affiliation` commands.

If affiliations have been listed before the first `\author` macro to ensure a particular ordering, be sure that any later `\affiliation` command for the given institution is an exact copy of the first, and also ensure that no ancillary data is given in these later instances.

A journal may determine the placement of these ancillary information footnotes. One journal may put all such footnotes at the start of the bibliography

while another journal may display them on the first page, as a footnote. One can override a journal style's default behavior by specifying explicitly the class option `bibnotes` (puts the footnotes at the start of the bibliography) or `nobibnotes` (puts them on the first page).

6.3.3 Specifying first names and surnames

Many authors have names in which either the surname appears first or in which the surname is made up of more than one name. To ensure that such names are accurately captured for indexing and other purposes, the `\surname` macro should be used to indicate which portion of a name is the surname. Similarly, there is a `\firstname` macro as well, although usage of `\surname` should be sufficient. If an author's surname is a single name and written last, it is not necessary to use these macros. These macros do nothing but indicate how a name should be indexed. Here are some examples;

```
% \author{Andrew \surname{Lloyd Weber}}
% \author{\surname{Mao} Tse-Tung}
```

6.4 The abstract

An abstract for a paper is specified by using the `abstract` environment:

```
% \begin{abstract}
% Text of abstract
% \end{abstract}
```

Note that with this package, the abstract must be specified before the `\maketitle` command, and there is no need to embed it in an explicit minipage environment.

6.5 PACS codes

Some authors are asked to supply suggested PACS codes with their submissions. The `\pacs` macro is provided as a way to do this:

```
\pacs{23.23.+x, 56.65.Dy}
```

The actual display of the PACS numbers below the abstract is controlled by two class options: `showpacs` and `noshowpacs`. In particular, this is now independent of the `preprint` option. `showpacs` must be explicitly included in the class options to display the PACS codes.

6.6 Keywords

A `\keywords` macro may also be used to indicate keywords for the article.

```
\keywords{nuclear form; yrast level}
```

This will be displayed below the abstract and PACS (if supplied). Like PACS codes, the actual display of the the keywords is controlled by two classoptions: `showkeys` and `noshowkeys`. An explicit `showkeys` must be included in the `\documentclass` line to display the keywords.

6.7 Institutional report numbers

Institutional report numbers can be specified using the `\preprint` macro. These will be displayed in the upper lefthand corner of the first page. Multiple `\preprint` macros may be supplied (space is limited though, so only three or less may actually fit).

6.8 maketitle

After specifying the title, authors, affiliations, abstract, PACS codes, and report numbers, the final step for formatting the front matter of the manuscript is to issue the `\maketitle` command:

```
\maketitle
```

This command must follow all of the macros listed above. The macro will format the front matter in accordance with the various class options that were specified in the `\documentclass` line (either implicitly through defaults or explicitly).

7 Compatability with LaTeX's Required Packages

Certain packages, usually ones written by members of the LaTeX Project itself, have been designated "required" and are distributed as part of standard LaTeX. These packages have been placed in a priviledged position vis á vis the LaTeX kernel in that they override the definitions of certain kernel macros.

The `ltxfront` package will be incompatible with any package that redefines any of the kernel macros that `ltxfront` patches—if that package is loaded *after* `ltxfront`.

Hereinafter follows some notes on specific LaTeX packages.

8 Implementation of package

8.1 Beginning of the package DOCSTRIP module

```
%<*package>
\def\package@name{ltxfront}%
\expandafter\PackageInfo\expandafter{\package@name}{%
 Title page macros for \protect\LaTeXe,
 by A. Ogawa (arthur_ogawa at sbcglobal.net)%
}%
```

8.2 Prerequisites

This package requires the use of the `ltxutil` package.

```
\let\class@name\package@name
\RequirePackage{ltxutil}%
%</package>
```

End of the preliminary portion of the package code.

8.3 Options

Package options are treated in this portion of the dtx file. This portion is guarded by the `options docstrip` code so that it can be merged into a larger package gracefully.

```
%<*options>
```

Package option turns on diagnostics that trace its operation. `\frontmatterverbose@sw` is set false by default: we do not elect to monitor the workings of these procedures.

```
\DeclareOption{frontmatterverbose}{\@booleantrue\frontmatterverbose@sw}%
\@booleanfalse\frontmatterverbose@sw
```

Package option inactivates its syntax, for testing purposes.

```
\DeclareOption{inactive}{\@booleanfalse\frontmatter@syntax@sw}%
\@booleantrue\frontmatter@syntax@sw
```

`\runinaddress@sw` The flag `\runinaddress@sw` signifies that author addresses are to be set on the same line with the author.

The `runinaddress` option is the only one that sets `\runinaddress@sw` to true.

FIXME: this option pertains to cases where `\groupauthors@sw` is true, but is independant of `\@affils@sw`. So, it should be possible to assert both `runinaddress` and `unsortedaddress` as well as each separately.

```
\@booleanfalse\runinaddress@sw
```

`\@affils@sw` If `\@affils@sw` is false, an address is never recognised as "new" and is therefore always entered into the affiliation list, stopping groups of authors at the same address being consolidated into the same list: the address will be printed the same number of times it is entered.

The `unsortedaddress` option is the only one that sets `\@affils@sw` to false.

```
\@booleantrue\@affils@sw
```

`\groupauthors@sw` The flag `\groupauthors@sw` signifies that authors are to be grouped, affecting the meaning of `\@author@present`.

Clients will want to set `\groupauthors@sw` true if they want grouped addresses or either of the following variants: for unsorted addresses: set `\@affils@sw` false; for run-in addresses, set `\runinaddress@sw` true.

```
\@booleanfalse\groupauthors@sw
```

`groupedaddress` If we take the `groupedaddress` option as the default, then we can look upon the `superscriptaddress` option as simply turning `\groupauthors@sw` to `\false@sw`.

`groupedaddress`, the default in APS journals, groups authors above a common address.

```
\DeclareOption{groupedaddress}{\clo@groupedaddress}%
\def\clo@groupedaddress{%
 \@booleantrue\groupauthors@sw
 \@booleantrue\@affils@sw
 \@booleanfalse\runinaddress@sw
}%
```

`unsortedaddress` `unsortedaddress` is similar to `groupedaddress`, but turns off `\@affils@sw`, with the result that each address that is entered will be printed.

```
\DeclareOption{unsortedaddress}{\clo@unsortedaddress}%
\def\clo@unsortedaddress{%
 \@booleantrue\groupauthors@sw
 \@booleanfalse\@affils@sw
 \@booleanfalse\runinaddress@sw
}%
```

`runinaddress` `runinaddress` is similar to `groupedaddress`, with the addition of the Boolean `\runinaddress@sw`, which causes the authors associated with a particular address group to be formatted in a paragraph instead of on separate lines.

```
\DeclareOption{runinaddress}{\clo@runinaddress}%
\def\clo@runinaddress{%
 \@booleantrue\groupauthors@sw
 \@booleantrue\@affils@sw
 \@booleantrue\runinaddress@sw
}%
```

`superscriptaddress` `superscriptaddress` presents author affiliations as superscripts. Authors with like affiliations share the same superscript.

FIXME: if `\groupauthors@sw` is false, would not `\@affils@sw` be a don't care?

```
\DeclareOption{superscriptaddress}{\clo@superscriptaddress}%
\def\clo@superscriptaddress{%
 \@booleanfalse\groupauthors@sw
 \@booleantrue\@affils@sw
 \@booleanfalse\runinaddress@sw
}%

%</options>
```

8.4 Process package options

```
%<*package>
\DeclareOption*{\OptionNotUsed}%
\ProcessOptions*
%</package>
```

9 Kernel of Front Matter

Here begins the kernel of the `frontmatter` package.

```
%<*kernel>
```

This package implements a new syntax for the title page of an article.

Special acknowledgment: this package uses concepts pioneered and first realized by William Baxter (mailto:web at superscript.com) in his SuperScript line of commercial typesetting tools, and which are used here with his permission.

These macros were first coded by David P. Carlisle for the American Physical society in the late 1990s as part of the development of REVTEX4. That development work was taken over by Arthur Ogawa, who is the author of this package.

9.1 Initialization

`\frontmatter@init` The `\frontmatter@init` procedure sets up all the registers for processing the title page. At present, this is done once, at `\class@documenthook` time. If multiple title pages are to be processed within a job, the initialization must be done for each.

```
\appdef\class@documenthook{\frontmatter@init}%
\let\frontmatter@init\@empty
```

9.2 The title command

`\title` `\do@title` The author uses `\title` to specify the title. As in the AMS classes, `\title` has an optional argument specifying a short form for use in running heads.

Once the `\title` is given, you can specify any combination of `\thanks`, `\email`, `\homepage`, and `\altaffiliation` commands, applying to the title of the document.

Compatibility note: as with all of the standard LaTeX commands that we override, we make here a private definition and later switch it in if so required.

```
\newcommand\frontmatter@title[2][]{%
 \def\@title{#2}%
 \def\@shorttitle{#1}%
 \let\@AF@join\@title@join
}%
```

`\@title` `\@shorttitle` The arguments to `\title` are saved in these internal macros, which are set up to produce a warning if `\title` has not been used before `\maketitle`.

Extensibility note: by using `\frontmatter@init` as the initialization procedure, we open the possibility of more than one title page per document.

```
\appdef\frontmatter@init{%
 \def\@title{\class@warn{No title}}%
 \let\@shorttitle\@empty
 \let\@title@aux\@title@aux@cleared
}%
```

\@title@join
\@title@join@
\@title@aux@cleared

```
\def\@title@join{\expandafter\@title@join@\@title@aux}%
\def\@title@join@#1#2{%
 \def\@title@aux{{\@join{\@separator}{#1}{#2}}}%
}%
\def\@title@aux@cleared{{}}%
```

9.3 The author, affiliation, and related commands

\c@affil
\c@collab

`\c@affil` stores the affiliation numbers used for the superscript marks. The `\newcounter` command defines `\theaffil`, which we never use: this counter is evaluated solely via `\the`.

`\c@collab` stores the collaboration numbers. The `\newcounter` command defines `\thecollab`. As with `\c@affil`, we never use `\thecollab`; only `\the \c@collab`.

Each counter must be cleared before frontmatter can be processed.

```
\newcounter{affil}%
\newcounter{collab}%
\appdef\frontmatter@init{%
 \c@affil\z@
 \c@collab\z@
}%
```

9.3.1 Commands for author or collaboration

\author

Each author is given in a *separate* `\author` command. This is similar to the AMS classes, but the AMS also try to support the `article` class's `\and` command for putting multiple authors in one `\author` command. This package, however, does not support the `\and` command.

When the `\author` command is encountered, the previous author, author group, affiliation, and affiliation group are at an end. This procedure moves the previous author to the author group (`\move@AU`), moves the previous affiliation to the affiliation group (`\move@AF`), and terminates the previous author/affiliation group (`\move@AUAF`) as the case may require.

Compatibility note: as with all of the standard LaTeX commands that we override, we make here a private definition and later switch it in if so required.

```
\newcommand\frontmatter@author{% implicit #1
 \@author@def{}% implicit #2
}%
```

\collaboration

Specify the collaboration (given after a group of authors).

This procedure does exactly what `\author` does, and sets a flag signifying that the `\collaboration` command was given.

Note: earlier versions attempted to ignore the collaboration command if it was considered inappropriate. We must not do this, however, because it throws the state machine out of kilter.

```
\def\collaboration{% implicit #1
```

```
 \@author@def{\@booleantrue\collaboration@sw}% implicit #2
}%
\appdef\frontmatter@init{%
 \@booleanfalse\collaboration@sw
}%
```

\@author@init
\@author@cleared
\@authorclear@sw
\@author@def
\@author@join@
\@author@join

Here are all the access procedures for the author data structure.

We define a flag value for \@author(private), a procedure \@author@gobble to read author data without any effect, a procedure for setting \@author to the flag value (\@author@init), a Boolean to test against the flag value (\@authorclear@sw), and a procedure to add new ancillary data to the author.

Note that expanding the functionality of the \collaboration command entailed changing the data structures, because now all authors must remember the collaboration they belong to. This change affects procedures \@author@cleared and \@author@gobble.

```
\def\@author@cleared{{}{}{}}%
\def\@author@gobble#1#2#3{}%
\def\@author@init{%
 \let\@author\@author@cleared
 \@booleanfalse\collaboration@sw
}%
\def\@authorclear@sw{\@ifx{\@author\@author@cleared}}%
\appdef\frontmatter@init{%
 \@author@init
}%

\def\@author@def#1#2{%
 \frontmatterverbose@sw{\typeout{\string\author\space\string\collaboration}}{}%
 \move@AU\move@AF\move@AUAF
 \let\@AF@join\@author@join
 #1%
 \def\@author{{#2}{}}%
}%
\def\@author@join@#1#2#3{%
  \def\@author{{#1}{\@join{\@separator}{#2}{#3}}}%
}%
\def\@author@join{\expandafter\@author@join@\@author}%
```

\move@AU
\add@AUCO@grp

The \move@AU procedure moves the most recent author to the current author group \AU@grp, or the most recent collaboration to the collaboration group \CO@grp. If we recently did a \@author@init, this is a no-op.

To accommodate the expanded syntax of \collaboration, we add a new field to those associated with an author (collaborator), namely the value of the collaboration counter.

```
\def\move@AU{%
 \@authorclear@sw{}{%
  \collaboration@sw{%
   \advance\c@collab\@ne
   \@argswap{\CO@grp\CO@opr}%
```

```
  }{%
   \@argswap{\AU@grp\AU@opr}%
  }%
   {%
    \expandafter\@argswap@val
    \expandafter{\@author}%
     {\expandafter\@argswap@val\expandafter{\the\c@collab}{\add@AUCO@grp}}%
   }%
 }%
 \@author@init
}%
```

The procedure `\add@AUCO@grp` adds to the specified group #3 using the given intercollated operator #4, with arguments {#1} and #2, where the first argument is the collaboration number, the second is the author name, and the third the joined attributes of the author. For example, `\move@AU` could finally execute: `\appdef \AU@grp {\AU@opr{1}{A. Author}{}}`.

```
\def\add@AUCO@grp#1#2#3#4{%
 \appdef#3{#4{#1}#2}%
 \frontmatterverbose@sw{\say#3}{}%
}%
```

`\@author@finish` We define a procedure to terminate author/affiliation parsing just before the title block is formatted. We detect the case where an author group is not followed by an affiliation group.

We first finish off any author- or affiliation groups in progress. Next, we detect the case where there is an author group or a collaboration group in progress, but no affiliation group. If so, we emit a diagnositic message and act as if a `\noaffiliation` command had been given. In any case, we finish off any author/affiliation pair that may remain. Formatting of the title block may now proceed.

```
\def\@author@finish{%
 \frontmatterverbose@sw{\typeout{\string\@author@finish}}{}%
 \move@AU\move@AF
 \@ifx{\AU@grp\@empty}{%
  \@ifx{\CO@grp\@empty}%
 }{%
  \false@sw
 }%
 {}{%
  \@ifx{\AF@grp\@empty}{%
   \begingroup
    \let\href\@secondoftwo
    \let\AU@opr\@secondofthree
    \let\CO@opr\@secondofthree
    \let\footnote\@gobble
    \@ifx{\CO@grp\@empty}{%
     \class@warn{Assuming \string\noaffiliation\space for authors}%
     \frontmatterverbose@sw{\say\AU@grp}%
```

```
    }{%
     \class@warn{Assuming \string\noaffiliation\space for collaboration}%
     \frontmatterverbose@sw{\say\CO@grp}{}%
    }%
   \endgroup
   \@affil@none\move@AF
  }{}%
 }%
 \move@AUAF
}%
\def\@secondofthree#1#2#3{#2}%
```

`\@join` The procedure `\@join` inserts a separator between two tokens, or, if the first token is nil, elides both that token and the separator.

```
\def\@join#1#2#3{%
  \@if@empty{#2}{#3}{#2#1#3}%
}%
\def\@separator{;\space}%
```

`\surname` `\firstname` No-op to allow better post-processing of the file.

```
\let\surname\@firstofone
\let\firstname\@firstofone
```

`\and` The original LaTeX idea of using one `\author` command to capture all authors and their address just doesn't work with multiple authors possibly sharing addresses, so in this class disable `\and`.

```
\newcommand\frontmatter@and{\class@err{\protect\and\space is not supported}}
```

9.3.2 Commands for affiliation

`\affiliation` After each group of authors with the same address, give that address in `\affiliation`. If later in the list you have an author with the address of an earlier author, repeat the `\affiliation` command: the system will detect the equivalence and (if using the `superscriptaddress` option) only print the affiliation once, reusing the superscript marker.

When the `\affiliation` command is encountered, the current author and author group (if any) are at an end. Also, the current affiliation is at an end.

FIXME: changes to catcode required elsewhere now?

```
\def\cat@comma@active{\catcode`\,\active}%
{\cat@comma@active\gdef,{\active@comma}}%
\def\active@comma{,\penalty-300\relax}%
\newcommand\affiliation{%
 \frontmatterverbose@sw{\typeout{\string\affiliation}}{}%
 \move@AU\move@AF
 \begingroup
  \cat@comma@active
  \@affiliation
}%
```

```
\def\@affiliation#1{%
 \endgroup
 \let\@AF@join\@affil@join
 \@affil@def{#1}%
}%
```

\noaffiliation

User-level command \noaffiliation signifies that the current group of authors, or the current collaboration, has *no* affiliation.

This is implemented by acting as if the \affiliation command has been given, and using a flag value of {\relax}, which adds an element to the affiliation group of this author, albeit a nil one (\AFF@opr {\relax} {}).

Note that this procedure does substantially the same as \@author@finish. (the latter, If it does not bail out altogether, differs chiefly in providing diagnostics).

```
\newcommand\frontmatter@noaffiliation{%
 \frontmatterverbose@sw{\typeout{\string\noaffiliation}}{}%
 \move@AU\move@AF
```

In \@author@finish, there is code testing if an effective \noaffiliation should be executed now. Here, we do so in any case.

```
 \@affil@none\move@AF
 \move@AUAF
}%
\def\blankaffiliation{{}}%
```

\@affil@cleared
\@affil@nil
\@affil@init
\@affil@none
\@affilclear@sw
\@affil@def
\@affil@join@
\@affil@join

Here are all the access procedures for the affiliation data structure. Note the similarity to those of the \@author data structure.

We define a flag value for \@affil (private), a procedure for setting \@affil to the flag value (\@affil@init), a Boolean to test against the flag value (\@affilclear@sw), a procedure \@affil@def to load a value into the structure, and a procedure to add new ancillary data to the affiliation.

```
\def\@affil@cleared{{{}}{}}%
\def\@affil@nil{{\relax}{}}%
\appdef\frontmatter@init{%
 \@affil@init
}%
\def\@affil@none{%
 \let\@affil\@affil@nil
}%
\def\@affil@init{%
 \let\@affil\@affil@cleared
}%
\def\@affilclear@sw{\@ifx{\@affil\@affil@cleared}}%
\def\@affil@def#1{%
 \def\@affil{{#1}{}}%
}%
\def\@affil@join@#1#2#3{%
  \def\@affil{{#1}{\@join{\@separator}{#2}{#3}}}%
}%
```

```
\def\@affil@join{\expandafter\@affil@join@\@affil}%
```

\move@AF Move the most recent affiliation to the affiliation group. If we recently did a \@affil@init, this is a no-op.

Set the \temp@sw to false, then execute the affiliation list (\@AFF@list) with the intercollated operator set to \@affil@match and with \@tempa holding the first component of \@affil, the current affiliation, which is to be matched.

If the current affiliation has been seen before, then by side effect \temp@sw will be set to true, \@tempc will be set to the matching affiliation number, and \@tempd will be set to the affiliation's ancillary data. The Boolean \@affils@sw being false prevents this test from ever returning a "true" result.

This procedure uses \@tempa and sets \@tempc, \@tempd, and \temp@sw.

```
\def\move@AF{%
 \@affilclear@sw{}{%
  \@booleanfalse\temp@sw
  \let\@tempd\@empty
  \@affils@sw{%
    \expandafter\@affil@addr@def\expandafter\@tempa\@affil
    \def\AFF@opr{\@affil@match\@tempa}%
    \@AFF@list
  }{}\temp@sw
```

True clause: This affiliation has been seen before. If ancillary data for the affiliation have been given, but are not identical to those seen before, give a warning.

```
  {%
   \expandafter\@affil@aux@def\expandafter\@tempb\@affil
   \@ifx{\@tempb\@empty}{}{%
    \@ifx{\@tempb\@tempd}{}{%
     \class@warn{%
      Ancillary information for \@tempa\space must not be different!
      Please put all of it on the first instance%
     }%
    }%
   }%
  }%
```

False clause: This affiliation is a new one: increment the affiliation counter to get a unique affiliation ID and append the new ID, collaboration number, address, and auxiliary information to the list \@AFF@list.

Note that, with the expanded syntax of \collaboration, we need to label each address with its associated collaboration.

Note also that the \noaffiliation case is handled by recognizing the flag value \@affil@nil.

```
  {%
   \@ifx{\@affil\@affil@nil}{%
    \def\@tempc{0}%
    \@argswap@val{0}%
   }{%
    \advance\c@affil\@ne
```

```
    \expandafter\def\expandafter\@tempc\expandafter{\the\c@affil}%
    \expandafter\@argswap@val\expandafter{\the\c@affil}%
   }%
   {%
    \expandafter\@argswap@val\expandafter{\the\c@collab}{%
     \expandafter\@argswap@val\expandafter{\@affil}{%
      \add@list@val@val@val\@AFF@list\AFF@opr
     }%
    }%
   }%
  }%
```

In either case, append this affiliation number to the `\AF@grp`, the affiliation group of the current author group. (It will later be considered for the `\@AFG@list`.) Then assign the flag value to `\@affil` that signifies it has been cleared.

```
  \appdef@eval\AF@grp\@tempc
  \frontmatterverbose@sw{\say\AF@grp}{}%
  \@affil@init
 }%
}%
\def\@affil@addr@def#1#2#3{%
 \def#1{#2}%
}%
\def\@affil@aux@def#1#2#3{%
 \def#1{#3}%
}%
```

Procedure `\add@list@val@val@val` appends to the given list #1 using the given intercollated operator #2, with arguments #5 (affiliation ID) and #4 (collaboration ID) delimited by braces, and #3 bare. In the end, `\move@AF` may execute something like `\appdef \@AFF@list \AFF@opr` {1} {3} {My Institution} {thanks}.

```
\def\add@list@val@val@val#1#2#3#4#5{%
 \appdef#1{#2{#5}{#4}#3}%
 \frontmatterverbose@sw{\say#1}{}%
}%
```

`\@affil@match` The procedure `\@affil@match` is the definition for `\AFF@opr`, the intercollated operator for `\@AFF@list`, when adding a new affiliation: it seeks a match with any earlier affiliation. When the affiliation list is executed, this procedure tests each entry against the 'new' affiliation (which has been stored in `\@tempa`) and its associated collaboration number (in `\@tempb`).

If `\groupauthors@sw` is true, then we require the collaboration ID to match, otherwise there is no such requirement, as is appropriate for the `superscriptaddress` class option.

Uses `\temp@sw`, `\@tempc`, and `\@tempd` to communicate back: if it returns `\temp@sw` true, a match was found. Then `\@tempc` will be the affiliation ID of the matching entry and `\@tempd` will be the auxiliary information of the matching entry.

```
\def\@affil@match#1#2#3#4#5{%
```

```
\temp@sw{}{%
 \def\@tempifx{#4}%
 \@ifx{\@tempifx#1}{%
  \groupauthors@sw{%
   \@ifnum{#3=\c@collab}{%
    \true@sw
   }{%
    \false@sw
   }%
  }{%
   \true@sw
  }%
 }{%
  \false@sw
 }%
 {%
  \@booleantrue\temp@sw
  \def\@tempc{#2}%
  \def\@tempd{#5}%
 }{%
 }%
}%
```

`\move@AUAF` Append `\AF@grp`, `\AU@grp`, and `\CO@grp` to `\@AAC@list`, append `\CO@grp` to `\@AFG@list`, and merge any new `\AF@grp` to `\@AFG@list`.

The entire procedure is predicated on something non-trivial to move.

If both author group `\AU@grp` and affiliation group `\AF@grp` are nil, bail out.

No, try that again:

If we have seen any affiliations (`\AF@grp` is not nil), then it is time to move the current affiliation group and author group to `\@AAC@list`. If not, we are picking up authors into an author group, and we should bail out.

To extend this scheme to the new `\collaboration` abilities, we must append an affiliation `\AF@opr` to `\@AAC@list` also in the case where there was no affiliation specified (e.g., `\noaffiliation`). This affiliation will be a nil affiliation: `\AF@opr{{0}}`.

FIXME: only one of `\CO@grp` or `\AU@grp` is non-empty at this point, but this code does not appear to recognize this fact!

```
\def\move@AUAF{%
 \frontmatterverbose@sw{\say\AU@grp\say\AF@grp\say\CO@grp}{}%
 \@ifx{\AF@grp\@empty}{%
    \@ifx{\@empty\CO@grp}{%
```

Both `\AF@grp` and `\CO@grp` are empty: nothing to do. But what if `\AU@grp` is not empty?

```
    }{%
```

Append the nil `\AF@opr` to `\@AAC@list`.

```
     \appdef     \@AAC@list{\AF@opr{{0}}}%
```

Append the collaboration group to the `\@AAC@list`; no intercollated operator need be added, because we already have `\CO@opr`.

```
     \appdef@e   \@AAC@list{\CO@grp}%
     \appdef@e   \@AFG@list{\CO@grp}%
     \let\CO@grp\@empty
    }%
}{%
```

Append current affiliation group (expansion of `\AF@grp`) to the author/affiliation list (`\@AAC@list`), using `\AF@opr` as the intercollated operator; also append the current authors (`\AU@grp`).

```
      \appdef     \@AAC@list{\AF@opr}%
      \appdef@eval\@AAC@list{\AF@grp}%
      \appdef@e   \@AAC@list{\AU@grp}%
```

Determine if either `\AU@grp` or `\CO@grp` is non-empty.

```
    \@ifx{\@empty\AU@grp}{%
     \@ifx{\@empty\CO@grp}%
    }{%
     \false@sw
    }%
```

At this point, there is a Boolean in TEX's scanner; it will parse the following two brace-delimited sequences and select one of them.

```
    {%
```

This branch is executed if and only if both `\AU@grp` and `\CO@grp` are empty.

```
    }{%
```

There was either a `\author` or a `\collaboration` command. Append the current affiliation group to `\@AFG@list`, (only if it is not already there) using `\AFG@opr` as the intercollated operator.

Note that `\@AFG@list` is a list of *unique* affiliation *groups*, and building it entails an N^2 computation.

```
      \@booleanfalse\temp@sw
      \def\AFG@opr{\x@match\AF@grp}%
      \let\CO@opr\@author@gobble
      \@AFG@list
      \temp@sw{}{%
        \appdef     \@AFG@list{\AFG@opr}%
        \appdef@eval\@AFG@list{\AF@grp}%
      }%
```

If the collaboration `\CO@grp` is not empty, we append it to the `\@AFG@list` and to the `\@AAC@list`.

```
     \@ifx{\@empty\CO@grp}{}{%
       \appdef@e   \@AAC@list{\CO@grp}%
       \appdef@e   \@AFG@list{\CO@grp}%
       \let\CO@grp\@empty
     }%
    }%
```

Finally, null out the macros holding the author group `\AU@grp`, affiliation group `\AF@grp`, and collaboration group `\CO@grp`.

```
    \let\CO@grp\@empty
    \let\AU@grp\@empty
    \let\AF@grp\@empty
}%
\frontmatterverbose@sw{\say\@AAC@list\say\@AFG@list}{}%
}%
```

`\AU@grp` `\AF@grp` `\@AAC@list` `\@AFG@list` `\@AFF@list`

The control sequence name `\AU@grp` accumulates consecutive `\author` entries (with `\AU@opr` as the intercollated operator) and `\collaboration` entries (with `\CO@opr` as the intercollated operator). Ultimately, its contents are appended to the author/affiliation list `\@AAC@list` and to the list of affiliation groups, `\@AFG@list`. It must be initialized to `\@empty`.

Note on `\AU@opr` and `\CO@opr`: it is essential to treat these two operators to the greatest extent possible on an equal footing. Therefore we invariably assign values to the pair of them within the same procedure.

The macro `\AF@grp` accumulates affiliation IDs into an affiliation group. It must be empty at the beginning of the frontmatter.

The `\@AAC@list` macro accumulates authors (using `\AU@opr` as the intercollated operator), collaborations (using `\CO@opr` as the intercollated operator), and affiliations (using `\AF@opr` as the intercollated operator).

It must be empty at the beginning of the frontmatter.

The macro `\@AFG@list` accumulates affiliation groups, with `\AFG@opr` as the intercollated operator, and collaborations, with no intercollated operator. This token list is employed when and only when `groupedaddress` is in effect.

The macro `\@AFF@list` accumulates the list of affiliations with `\AFF@opr` as the intercollated operator. It must be empty at the beginning of the frontmatter.

Each `\affiliation` command in the document contributes to this list, as long as the argument of that instance of the command is "new".

Each entry in this list consists of the `\AFF@opr` intercollated operator followed by three brace-delimited tokens, representing, in order:

1. the affiliation's unique ID, a number increasing monotonically for each new entry,
2. the optional argument of the `\affiliation` command, the footnote to the affiliation, and
3. the text of the affiliation.

At various junctures in the code, the control sequence name `\AFF@opr` assumes the meaning of `\@affil@group`, `\affil@script`, or is defined to expand to `\@affil@match\@tempa`, and the list `\@AFF@list` is expanded.

For each of these lists, here is the syntax of an element and the usage of the associated operator:

```
%\@AFF@list \AFF@opr{⟨affiliation ID⟩}{⟨collaboration ID⟩}{⟨address⟩}{⟨auxiliary info⟩} \frontmatt
%\@AFG@list \AFG@opr{{⟨affiliation ID⟩}..{⟨affiliation ID⟩}} \frontmatter@author@produce@group
%\@AAC@list \AF@opr{{⟨affiliation ID⟩}..{⟨affiliation ID⟩}} (\AU@opr|\CO@opr){⟨collaboration ID⟩}{
%
%\AFF@opr \@affil@match\@tempa \@affil@script \@affil@group
%\AFG@opr \affils@present@group \x@match\AF@grp
%\AF@opr \@gobble \@affilID@def \@affilID@count\AF@temp\@tempcnta \@affilID@match\AF@temp
%\AU@opr \@secondofthree \@author@count\@tempcnta \@author@present \@author@gobble \@author@count
%\CO@opr \@secondofthree \@collaboration@count\AU@temp\@tempcnta \@collaboration@present\AU@temp
%
%\CO@opr\@author@cleared
%
```

```
\appdef\frontmatter@init{%
 \let\AU@grp\@empty
 \let\CO@grp\@empty
 \let\AF@grp\@empty
 \let\@AAC@list\@empty
 \let\@AFG@list\@empty
 \let\@AFF@list\@empty
}%
```

`\@AF@join` The `\csname` `\@AF@join` is a procedure used within `\email`, `\homepage`, `\thanks`, and `\altaffiliation`. When such a command appears after an `\author`, `\collaboration`, or `\affiliation`command, the `\@AF@join` procedure appends the argument given to the appropriate macro. Its default is to give an error message, since these commands are legal only within the particular context mentioned.

```
\appdef\frontmatter@init{%
 \let\@AF@join\@AF@join@error
}%
\def\@AF@join@error#1{%
 \class@warn{%
  \string\email, \string\homepage, \string\thanks, or \string\altaffiliation\space
  appears in wrong context.
 }%
}%
```

9.3.3 Commands for auxiliary information

`\email` Just tacks the email address on to the current author or affiliation.

```
\def\sanitize@url{%
 \@makeother\%%
 \@makeother\~%
 \@makeother\_%
}%
\newcommand*\email[1][]{\begingroup\sanitize@url\@email{#1}}%
\def\@email#1#2{%
 \endgroup
 \@AF@join{#1\href{mailto:#2}{#2}}%
}%
```

\homepage Just tacks the URL on to the current author or affiliation. Note: group opened in \homepage is closed in \@homepage.

```
\newcommand*\homepage[1][]{\begingroup\sanitize@url\@homepage{#1}}%
\def\@homepage#1#2{%
 \endgroup
 \@AF@join{#1\href{#2}{#2}}%
}%
```

\href The document may load a package that defines either of these commands; if not, we give a default meaning.

```
\appdef\class@documenthook{%
 \providecommand\href[1]{}%
}%
```

\thanks The operative version of \thanks appends an item to \@author, or \@affil's auxiliary data.

```
\def\frontmatter@thanks{% implicit #1
  \@AF@join
}%
```

\altaffiliation Implemented more or less like \thanks but shares the affiliation counter. Optional argument may be used to give explanatory text eg 'currently staying at' This will be placed before the address, if used in a footnote.

```
\newcommand*\altaffiliation[2][]{%
  \@AF@join{#1#2}%
}%
```

9.3.4 Procedures for author, collaboration, and affiliation

Macros that list off, say, authors, and which require punctuation, like "A, B, and C", will use \set@listcomma@list, \set@listcomma@count, \@listand, and \@listcomma. These macros use the \@tempcnta register to keep track of how many items remain to be listed off.

At present, \@author@present will decrement that counter.

\set@listcomma@list
\set@listcomma@count Pop the author count for this collaboration.

Note: by side effect, it assigns \@listcomma: for a list of length two, suppress comma addition.

```
\def\set@listcomma@list#1{%
  \expandafter\@reset@ac\expandafter#1#1{0}\@reset@ac{%
   \let\@listcomma\relax
  }{%
   \let\@listcomma\@listcomma@comma
  }%
}%
\def\set@listcomma@count#1{%
  \@ifnum{#1=\tw@}{%
    \let\@listcomma\relax
```

```
  }{%
    \let\@listcomma\@listcomma@comma
  }%
}%
```

Does the actual pop operation, then generates a Boolean which selects one of the two assignments for `\@listcomma`.

Note: this procedure sets `\@tempcnta` to reflect the number of items to list off.

```
\def\@reset@ac#1#2#3\@reset@ac{%
  \def#1{#3}%
  \@tempcnta#2\relax
  \@ifnum{#2=\tw@}%
}%
```

`\listand` Might need extending with penalties etc.

Note: this procedure expects `\@tempcnta` to reflect the current number of items remaining to list off.

```
\def\@listand{\@ifnum{\@tempcnta=\tw@}{\andname\space}{}}%
```

`\@listcomma` This definition, with `\@ne` puts a comma before and.

David Carlisle says: In UK English (at least) would have `\tw@` here, which would then implement the convention of indicating conjunction with comma or "and", but not both.

Note: this procedure expects `\@tempcnta` to reflect the current number of items remaining to list off.

```
\def\@listcomma@comma{\@ifnum{\@tempcnta>\@ne}{,}{}}%
\def\@listcomma@comma@UK{\@ifnum{\@tempcnta>\tw@}{,}{}}%
```

`\@collaboration@gobble` This command will be the synonym of `\CO@grp` during the expansion of `\@AAC@list`, within `\affils@present@group`execution (in a `groupedaddress` situation).

```
\def\@collaboration@gobble#1#2#3{}%
```

`\doauthor` Main control over how authors are typeset.

`#1` is loaded by `\author`

`#2` is loaded by `\email`, `\homepage`, `\altaffiliation`, or `\thanks`

`#3` is the superscript affiliation, if at all used.

First, the author name is formatted, followed by a comma, then come any marks relating to affiliation (if present), then come the homepage URL and email address, if any, with appropriate punctuation.

Here, as in `\@affil@script`, the parameter `\affil@cutoff` controls whether the argument #3 is produced as a superscript label connecting this author with its affiliation. (If there is only one affiliation on the title page, it makes sense to remove the superscript.)

Note that argument #3 must effectively execute either `\aftergroup\true@sw` or `\aftergroup\false@sw` (exactly once!). In the case of `superscriptaddress`, this is done when `\@author@present@script` invokes `\doauthor` with its third argument containing `\@affil@present@script`. Otherwise (e.g., `groupedaddress`),

\doauthor is invoked by \@author@present@group with its third argument containing \@affil@present@group.

```
\def\doauthor#1#2#3{%
  \ignorespaces#1\unskip\@listcomma
  \begingroup
   #3%
```

At this point, we must have queued up a Boolean (either \true@sw or \false@sw).

```
  \@if@empty{#2}{\endgroup{}{}}{\endgroup{\comma@space}{}\frontmatter@footnote{#2}}%
  \space \@listand
}%
```

\x@match Procedure \x@match is used as an alias \AFG@opr, probing the list of affiliation groups and setting a Boolean if a match is found to #1 (usually the current affiliation group).

Procedure \y@match is used as an alias to \@TBN@opr, probing the list of unique title block footnotes for a match to the footnote text: when the first match is found it sets a Boolean, defining \@tempb to the number of that footnote.

```
\def\x@match#1#2{%
 \temp@sw{}{%
  \def\@tempifx{#2}%
  \@ifx{\@tempifx#1}{%
    \@booleantrue\temp@sw
  }{%
  }%
 }%
}%
\def\y@match#1#2#3{%
 \temp@sw{}{%
  \def\@tempifx{#3}%
  \@ifx{\@tempifx#1}{%
    \@booleantrue\temp@sw
    \def\@tempb{#2}%
  }{%
  }%
 }%
}%
```

\frontmatter@footnote Used when typesetting a title block footnote. (Item 290: merge duplicates.)

Maintains a list of frontmatter footnotes along with associated device. Uniquifies each footnote by traversing the list and looking for a match to the footnote text. If found, define \@tempb to the corresponding footnote device, and typeset it with \@footnotemark. If never found, manually increment footnote counter, determine the corresponding footnote device, and add footnote text and device to the list of frontmatter footnotes.

The list \@FMN@list is processed later in the title block production.

Note that this method of making footnotes runs afoul of hyperref's patches of the LaTeX kernel. Therefore, we avoid trouble by refraining from using \@footnotemark and by creating our own hypertext links.

```
\def\frontmatter@footnote#1{%
 \begingroup
  \@booleanfalse\temp@sw
  \def\@tempa{#1}%
  \let\@tempb\@empty
  \def\@TBN@opr{\y@match\@tempa}%
  \@FMN@list
```

By side effect, `\@tempb` holds the value of the footnote counter if This note has been seen before. Otherwise `\temp@sw` is false.

```
  \temp@sw{%
   \expandafter\frontmatter@footnotemark
   \expandafter{\@tempb}%
  }{%
```

Execute `\frontmatter@foot@mark`, feeding it the value of the current footnote counter. This call requires two levels of pre-expansion to accomplish, the first to expand `\csname`, the second to evaluate `\the`.

```
   \stepcounter\@mpfn
   \expandafter\expandafter
   \expandafter\frontmatter@foot@mark
   \expandafter\expandafter
   \expandafter{%
   \expandafter \the\csname c@\@mpfn\endcsname
                }{#1}%
  }%
 \endgroup
}%
```

Procedure `\frontmatter@foot@mark` lays down the footnote device (mark) and takes care of the footnote text. The latter merely involves queueing that text (along with its associated device) into `\@FMN@list`.

```
\def\frontmatter@foot@mark#1#2{%
 \frontmatter@footnotemark{#1}%
 \g@addto@macro\@FMN@list{\@TBN@opr{#1}{#2}}%
}%
\appdef\frontmatter@init{%
 \global\let\@FMN@list\@empty
}%

\def\frontmatter@footnotemark#1{%
 \leavevmode
 \ifhmode\edef\@x@sf{\the\spacefactor}\nobreak\fi
  \begingroup
   \hyper@linkstart {link}{frontmatter.#1}%
    \csname c@\@mpfn\endcsname#1\relax
    \def\@thefnmark{\frontmatter@thefootnote}%
    \@makefnmark
   \hyper@linkend
  \endgroup
 \ifhmode\spacefactor\@x@sf\fi
```

```
 \relax
}%
```

9.4 The keywords command

\keywords
\@keywords

Usual game, save text in a macro for processing by \maketitle.

```
\def\keywords#1{%
  \aftermaketitle@chk{\keywords}%
  \gdef\@keywords{#1}%
}%
\appdef\frontmatter@init{%
 \let\@keywords\@empty
}%
```

9.5 The \date command and related commands

\date
\received
\revised
\accepted
\published
\@date
\@received
\@revised
\@accepted
\@published

Use the \date command to specify the document date, the \received command to specify the date received, \revised for date revised, \accepted for date accepted, and \published for date published. Normally only \date to be used by author, the remainder used only by editors.

DPC: As for \keywords, but this time don't flag a warning if there is no revision date specified.

In each case the user-level command defines the value of a macro which buffers the data entered by the user. For instance, \received causes \@received to acquire a value. The optional argument allows the user to override the text that will be typeset along with the date; the default value of that text is itself a localized macro.

```
\newcommand*\frontmatter@date[2][\Dated@name]{\def\@date{#1#2}}%
\def\@date{}%
\newcommand*\received[2][\Received@name]{\def\@received{#1#2}}%
\def\@received{}%
\newcommand*\revised[2][\Revised@name]{\def\@revised{#1#2}}%
\def\@revised{}%
\newcommand*\accepted[2][\Accepted@name]{\def\@accepted{#1#2}}%
\def\@accepted{}%
\newcommand*\published[2][\Published@name]{\def\@published{#1#2}}%
\def\@published{}%
```

9.6 The pacs command

PACS, the Physics and Astronomy Classification Scheme.

\pacs
\@pacs

As for \keywords

```
\def\pacs#1{%
  \aftermaketitle@chk{\pacs}%
  \gdef\@pacs{#1}%
}%
\appdef\frontmatter@init{%
```

```
 \let\@pacs\@empty
}%
```

9.7 The \preprint command

```
\def\preprint#1{\gappdef\@preprint{\preprint{#1}}}%
\appdef\frontmatter@init{%
 \let\@preprint\@empty
}%
```

9.8 The abstract environment

\absbox

```
\newbox\absbox
```

abstract

Abstract, as in AMS classes, must be specified *before* \maketitle. It just saves everything up in \absbox. Note that the specifics of how the abstract is to be formatted are set by \frontmatter@abstractwidth, \frontmatter@abstractheading, and \frontmatter@abstractfont.

Here we wish to set the abstract into type but save it away in a box, much like the \minipage command does.

Note that the \endabstract portion of the environment begins with code that mostly duplicates \endminipage, but without the \@iiiparbox. At the end, we simply transfer the contents of the \minipage into a box of our own.

However, we also have to end the gratuitous paragraph begun by \minipage. We ensure that no trace of that paragraph is left, by absorbing the \parindent box. Doing so destroys the paragraph entirely, except for the \parskip glue, and that morsel is pruned, because we are at the top of a page.

This mechanism is vulnerable, however, because anything placed into the horizontal list after the \parindent box will confound it. And that is exactly what happens under MikTEX: There seems to be an extension to the TEX standard operating here which inserts a \special at the beginning of every paragraph. (The mechnism is *not* based on \everypar.)

To accomodate this state of affairs, we have a new mechanism that offers an even more complete rewrite of minipage.

Note that an entry is made in the toc for the abstract; this has the side effect of creating a PDF bookmark for this element if the hyperref package is in effect.

```
\def\toclevel@abstract{1}%
\def\addcontents@abstract{%
 \phantomsection
 \expandafter\def\csname Parent0\endcsname{section*.2}%
 \expandafter\@argswap@val\expandafter{\abstractname}{\addcontentsline{toc}{abstract}}%
}%
\newenvironment{frontmatter@abstract}{%
  \aftermaketitle@chk{\begin{abstract}}%
  \global\setbox\absbox\vbox\bgroup
   \color@begingroup
```

```
    \columnwidth\textwidth
    \hsize\columnwidth
    \@parboxrestore
```

The following line switches footnotes to the `mpfoootnote` series. This action should rather be taken by `\frontmatter@abstractfont`, which can be customized. There is also an interaction with `\mini@note`.

```
    \def\@mpfn{mpfootnote}\def\thempfn{\thempfootnote}\c@mpfootnote\z@
    \let\@footnotetext\frontmatter@footnotetext
    \minipagefootnote@init
    \let\set@listindent\set@listindent@
    \let\@listdepth\@mplistdepth \@mplistdepth\z@
    \let@environment{description}{frontmatter@description}%
    \@minipagerestore
    \@setminipage
     \frontmatter@abstractheading
     \frontmatter@abstractfont
     \let\footnote\mini@note
     \expandafter\everypar\expandafter{\the\everypar\addcontents@abstract\everypar{}}%
}{%
     \par
     \unskip
     \minipagefootnote@here
     \@minipagefalse   %% added 24 May 89
     \color@endgroup
   \egroup
}%
```

`\frontmatter@footnotetext` We reimplement `\@mpfootnotetext` under a new name (so as not to be overridden by the `hyperref` package) and extend it to accomodate hyperrefs.

Note that this procedure is very like `ltxutil`'s `\mp@footnotetext`, except that it uses `\frontmatter@makefntext` instead of `\@makefntext`.

```
\long\def\frontmatter@footnotetext#1{%
  \minipagefootnote@pick
    \set@footnotefont
    \set@footnotewidth
    \@parboxrestore
    \protected@edef\@currentlabel{\csname p@\@mpfn\endcsname\@thefnmark}%
    \color@begingroup
      \frontmatter@makefntext{%
        \rule\z@\footnotesep\ignorespaces#1\@finalstrut\strutbox\vadjust{\vskip\z@skip}%
      }%
    \color@endgroup
  \minipagefootnote@drop
}%
```

`\ltx@no@footwarn`

```
\def\ltx@no@footnote{%
 \let\ltx@xfootnote\ltx@no@xfootnote\let\ltx@yfootnote\ltx@no@yfootnote
```

```
 \let\ltx@xfootmark\ltx@no@xfootmark\let\ltx@yfootmark\ltx@no@yfootmark
 \let\ltx@xfoottext\ltx@no@xfoottext\let\ltx@yfoottext\ltx@no@yfoottext
}%
\def\ltx@no@xfootnote[#1]#2{\ltx@no@footwarn\footnote}%
\def\ltx@no@yfootnote#1{\ltx@no@footwarn\footnote}%
\def\ltx@no@xfootmark[#1]{\ltx@no@footwarn\footnotemark}%
\def\ltx@no@yfootmark{\ltx@no@footwarn\footnotemark}%
\def\ltx@no@xfoottext[#1]#2{\ltx@no@footwarn\footnotetext}%
\def\ltx@no@yfoottext#1{\ltx@no@footwarn\footnotetext}%
\def\ltx@no@footwarn#1{%
 \class@warn{%
  The \string#1\space command is not legal on the title page;
  using \string\thanks\space instead might suit you: consult the manual for details%
 }%
}%
```

`\frontmatter@abstractheading` The default abstract head; journals will override this procedure.

```
\def\frontmatter@abstractheading{%
 \begingroup
  \centering\large
  \abstractname
  \par
 \endgroup
}%
```

`\frontmatter@abstractfont` The default type specification for the body of the abstract. Journals will override this setting.

```
\def\frontmatter@abstractfont{}%
```

`frontmatter@description` Within the abstract, the description environment is defined as follows:

```
\newenvironment{frontmatter@description}{%
 \list{}{%
  \leftmargin\z@
  \labelwidth\z@

 % \itemindent-\leftmargin
 %

  \itemindent\z@
  \let\makelabel\frontmatter@descriptionlabel
 }%
}{%
 \endlist
}%
\def\frontmatter@descriptionlabel#1{%
 \hspace\labelsep
 \normalfont\bfseries
 #1:%
}%
```

\frontmatter@abstractwidth The default setting is the full text width; journals can override this setting.

```
\def\frontmatter@abstractwidth{\textwidth}
```

\frontmatter@abstract@produce This procedure determines how the abstract is incorporated into the title block. We split this out in anticipation of the need to lift the limitation that the abstract not break over pages.

If we are in preprint style, we provide an easy pagebreak point immediately above the abstract. This means that the abstract will either fit entirely on the same page as the title block, or it will start a page of its own (which may itself break onto a subsequent page if necessary).

```
\def\frontmatter@abstract@produce{%
  \par
  \preprintsty@sw{%
   \do@output@MVL{%
    \vskip\frontmatter@preabstractspace
    \vskip200\p@\@plus1fil
    \penalty-200\relax
    \vskip-200\p@\@plus-1fil
   }%
  }{%
   \addvspace{\frontmatter@preabstractspace}%
  }%
   \begingroup
    \dimen@\baselineskip
    \setbox\z@\vtop{\unvcopy\absbox}%
    \advance\dimen@-\ht\z@\advance\dimen@-\prevdepth
    \@ifdim{\dimen@>\z@}{\vskip\dimen@}{}%
   \endgroup
   \begingroup
    \prep@absbox
    \unvbox\absbox
    \post@absbox
   \endgroup
  \@ifx{\@empty\mini@notes}{}{\mini@notes\par}%
  \addvspace\frontmatter@postabstractspace
}%
\appdef\frontmatter@init{\let\mini@notes\@empty}%
\let\prep@absbox\@empty
\let\post@absbox\@empty
```

\frontmatter@preabstractspace Space above and space below abstract in title block

\frontmatter@postabstractspace

```
\def\frontmatter@preabstractspace{.5\baselineskip}
\def\frontmatter@postabstractspace{.5\baselineskip}
```

9.9 Formatting the title block

titlepage This is LaTeX 2ε's `article` class version, with modifications.

```
\newenvironment{frontmatter@titlepage}{%
```

```
      \twocolumn@sw{\onecolumngrid}{\newpage}%
      \thispagestyle{titlepage}%
      \setcounter{page}\@ne
}{%
     \twocolumn@sw{\twocolumngrid}{\newpage}%
     \twoside@sw{}{%
        \setcounter{page}\@ne
     }%
}%
```

`\maketitle` Put it all together to format the title block.

Gotcha! If you expand `\csname`undefined`\endcsname`, you change the meaning of `\undefined` from "Undefined" to `\relax`. Watchout!

```
\def\frontmatter@maketitle{%
  \@author@finish
  \title@column\titleblock@produce
  \suppressfloats[t]%
```

Now save some memory.

```
  \let\and\relax
  \let\affiliation\@gobble
  \let\author\@gobble
  \let\@AAC@list\@empty
  \let\@AFF@list\@empty
  \let\@AFG@list\@empty
  \let\@AF@join\@AF@join@error
  \let\email\@gobble
  \let\@address\@empty
  \let\maketitle\relax
  \let\thanks\@gobble
  \let\abstract\@undefined\let\endabstract\@undefined
  \titlepage@sw{%
   \vfil
   \clearpage
  }{}%
}%
```

`\maketitle@Hy` We provide the means to take back the definition of `\maketitle` from `hyperref`: we do not need its help in providing hypertext services in the title page.

```
\def\maketitle@Hy{%
  \let\Hy@saved@footnotemark\@footnotemark
  \let\Hy@saved@footnotetext\@footnotetext
  \let\@footnotemark\H@@footnotemark
  \let\@footnotetext\H@@footnotetext
  \@ifnextchar[%]
   \Hy@maketitle@optarg
   {%
    \HyOrg@maketitle
    \Hy@maketitle@end
   }%
```

```
}%
\appdef\class@documenthook{%
  \@ifx{\maketitle\maketitle@Hy}{%
   \class@info{Taking \string\maketitle\space back from hyperref}%
   \let\maketitle\frontmatter@maketitle
  }{%
  }%
}%
```

`\titleblock@produce` This procedure produces the title block.

It effectively executes inside a group and always returns us to vertical mode.

Note: we assume that it is OK to set the footnote counter to zero at this point. Is this a safe assumption?

```
\def\titleblock@produce{%
 \begingroup
```

First, we restore the footnote mechanism to its default state, then we customize the way the footnote mark is produced and how it is formatted.

```
  \ltx@footnote@pop
```

The following three lines establish an independent footnote counter for use in the title block.

```
  \def\@mpfn{mpfootnote}%
  \def\thempfn{\thempfootnote}%
  \c@mpfootnote\z@
  \let\@makefnmark\frontmatter@makefnmark
  \frontmatter@setup
  \thispagestyle{titlepage}\label{FirstPage}%
```

Produce the title:

```
  \frontmatter@title@produce
```

Produce the author list:

```
  \groupauthors@sw{%
   \frontmatter@author@produce@group
  }{%
   \frontmatter@author@produce@script
  }%
```

Produce the dates:

```
  \frontmatter@RRAPformat{%
   \expandafter\produce@RRAP\expandafter{\@date}%
   \expandafter\produce@RRAP\expandafter{\@received}%
   \expandafter\produce@RRAP\expandafter{\@revised}%
   \expandafter\produce@RRAP\expandafter{\@accepted}%
   \expandafter\produce@RRAP\expandafter{\@published}%
  }%
```

Produce the abstract, PACS, and keywords, and end any paragraph.

```
  \frontmatter@abstract@produce
  \@ifx@empty\@pacs{}{%
```

```
  \@pacs@produce\@pacs
 }%
 \@ifx@empty\@keywords{}{%
  \@keywords@produce\@keywords
 }%
 \par
 \frontmatter@finalspace
\endgroup
}%
```

`\frontmatter@title@produce` The default formatting procedure for the article title. This procedure should take care of the vertical space below the title.

Clients may override this procedure, but it is more likely that they will use the hooks `\frontmatter@title@above`, `\frontmatter@title@format`, and `\frontmatter@title@below`.

```
\def\toclevel@title{0}%
\def\frontmatter@title@produce{%
 \begingroup
  \frontmatter@title@above
  \frontmatter@title@format
  \@title
  \unskip
  \phantomsection\expandafter\@argswap@val\expandafter{\@title}{\addcontentsline{toc}{title}}%
  \@ifx{\@title@aux\@title@aux@cleared}{}{%
   \expandafter\frontmatter@footnote\expandafter{\@title@aux}%
  }%
  \par
  \frontmatter@title@below
 \endgroup
}%
\appdef\let@mark{\let\\\relax}%
```

`\frontmatter@title@above` `\frontmatter@title@format` `\frontmatter@title@below` The default values for formatting specifications for the article title.

The procedure `\frontmatter@title@above` should take care of the vertical space above the title; `\frontmatter@title@below` below. The procedure `\frontmatter@title@format` should invoke any font switches, etc. that may apply to the title.

Journals will override this procedure.

```
\def\frontmatter@title@above{}%
\def\frontmatter@title@format{}%
\def\frontmatter@title@below{\addvspace{\baselineskip}}%
```

9.9.1 Authors and affiliations in superscriptaddress style

`ontmatter@author@produce@script` `\frontmatter@author@produce@script` is an alias for `\frontmatter@author@produce`, the procedure that formats the author/affiliation list.

In this case, the authors affiliations are being superscripted (class option `superscriptaddress`).

This procedure must ensure that any paragraph that it starts gets ended finally.

```
\def\frontmatter@author@produce@script{%
  \begingroup
    \let\@author@present\@author@present@script
    \frontmatterverbose@sw{\typeout{\string\frontmatter@author@produce@script:}\say\@AAC@list\s
```

When the author/affiliation list `\@AAC@list` is executed, `\@tempcnt` counts each author, and `\@tempa` stores a list of author indices for the current collaboration.

Note: this procedure uses `\AU@temp` to hold a list of author counts for each collaboration. Note: also uses `\@tempcnta` to communicate between procedures. This usage ot `\@tempcnta` is OK, because the expansion of `\@AAC@list` will trigger only the execution of our own procedures.

```
    \let\AU@temp\@empty
    \@tempcnta\z@
```

We wish to know how many authors are in each collaboration, for the purposes of `\listcomma` and `\listand`.

Here we assign values for the intercollated operators appearing within `\@AAC@list`, then execute that macro, registering the authors in each collaboration. Afterwards, clean up by emulating an empty collaboration.

The result, stored in `\AU@temp`, is a list of brace-delmited tokens, each a number representing the number of authors in that collaboration.

```
    \let\AF@opr \@gobble
    \def\AU@opr{\@author@count\@tempcnta}%
    \def\CO@opr{\@collaboration@count\AU@temp\@tempcnta}%
    \@AAC@list
    \expandafter\CO@opr\@author@cleared
```

Next, present the authors and their affiliations, with the collaborations interleaved.

We assign values for the intercollated operators appearing within `\@AAC@list`. Next, assign `\@listcomma` based on the number of authors in the first collaboration. Then expand `\@AAC@list`.

Upon completion, there will be two space tokens following the last author because of `\listand` processing; remove them, then end the last paragraph.

FIXME: this juncture would be a good time to effectively eliminate the unused affiliations in `\@AFF@list`.

```
    \begingroup
     \frontmatter@authorformat
     \let\AF@opr \@affilID@def
     \let\AU@opr \@author@present
     \def\CO@opr{\@collaboration@present\AU@temp}%
     \set@listcomma@list\AU@temp
     \@AAC@list
     \unskip\unskip
     \par
    \endgroup
```

We now list out the affiliations in the order they appeared.

```
    \begingroup
     \frontmatter@above@affiliation@script
     \let\AFF@opr \@affil@script
     \@AFF@list
     \frontmatter@footnote@produce
     \par
    \endgroup
  \endgroup
}%
```

`\@author@count` This version of `\AU@opr` counts the number of authors it processes.

```
\def\@author@count#1{%
 \advance#1\@ne
 \@author@gobble
}%
```

`\@collaboration@present` Format a collaboration. Note that we immediately end the paragraph thus begun, because we only support `superscriptaddress`-style processing.

Note also that the execution of the production procedures takes place inside of a group; for reasons of protective programming.

Bug note: originally, the number of authors in each collaboration was stored in `\@tempa`, but this exposed us to procedures in our production that modified the meaning of `\@tempa`.

The last action done in this procedure is to assign `\@listcomma` based on the number of authors in the next collaboration.

```
\def\@collaboration@present#1#2#3#4{%
 \par
 \begingroup
  \frontmatter@collaboration@above
  \@affilID@def{}%
  \@tempcnta\z@
  \@author@present{}{(\ignorespaces#3\unskip)}{#4}%
  \par
 \endgroup
 \set@listcomma@list#1%
}%
\def\frontmatter@collaboration@above{}%
```

`\@collaboration@count` Simply register each author in this collaboration. Note: #1 is a `\csname` to hold the value, #2 is the `\count` register holding the count of interest.

```
\def\@collaboration@count#1#2{%
 \appdef@eval#1{\the#2}#2\z@
 \@author@gobble
}%
```

`\@affilID@def` The `\@affilID@def` procedure, an alias of `\AF@opr` used during `\frontmatter@author@produce@script` processing, memorizes the affiliation in `\@affilID@temp`.

```
\def\@affilID@def{\def\@affilID@temp}%
```

`\@affilID@temp` The macro `\@affilID@temp` is used to communicate between `\doauthor` and instances of `\AF@opr` within the author list.

```
\let\@affilID@temp\@empty
```

`\affils@present@script` An alias of `\AFG@opr`, this procedure is applied to elements in `\@AFG@list`. It builds `\@AFU@list`, a list of unique affiliations found within affiliation groups.

```
%\def\affils@present@script#1{%
% \get@affil#1{}%
%}%
%\def\get@affil#1{%
% \def\@tempa{#1}%
% \@ifx{\@empty\@tempa}{}{%
%  \@booleanfalse\temp@sw
%  \def\AF@opr{\x@match\@tempa}%
%  \@AFU@list
%  \temp@sw{}{%
%   \appdef\@AFU@list{\AF@opr{#1}}%
%  }%
%  \get@affil
% }%
%}%
%
```

`\affil@present@script` An alias of `\AF@opr`, this procedure is applied to `\@AFU@list` when formatting the affiliations under `superscriptaddress`. It in turn applies `\affil@script` to `\@AFF@list`, thereby formatting the affiliation associated with this affiliation index.

Note: it traverses the `\@AFF@list` for each entry in the `\@AFU@list`, thereby making this portion of code execute in N^2 time.

```
%\def\affil@present@script#1{%
% \def\@tempa{#1}%
% \begingroup
%  \frontmatter@affiliationfont
%  \let\AFF@opr \affil@script
%  \@AFF@list
% \endgroup
%}%
%
```

`\affil@script` Alias of `\AFF@opr`, this procedure is applied to `\@AFF@list` when formatting the affiliations under `superscriptaddress`. It rejects all entries not matching the affiliation index held in `\@tempa`.

```
\def\affil@script#1#2#3{%
 \def\@tempifx{#1}\@ifx{\@tempifx\@tempa}{%
  \@if@empty{#2}{}{%
   \par
   \begingroup
```

```
    \def\@thefnmark{#1}\@makefnmark\ignorespaces
    #2%
    \@if@empty{#3}{}{\frontmatter@footnote{#3}}%
    \par
   \endgroup
  }%
 }{}%
}%
```

`\@affil@script` Alias of `\AFF@opr`: The affiliations are being formatted in the case where affiliations are being superscripted: make a list out of the affiliations with the numbers printed. Society-specific code can change the formatting by overriding the definition of `\frontmatter@affiliationfont`.

A client may choose to set `\affil@cutoff` to either `\@ne` (the default) or `\tw@`. The latter setting will enforce a rule to the effect that if a single affiliation appears on the title page, then no affiliation superscript at all is produced.

`#1`—Affiliation ID: a number; zero signifies a `\noaffiliation`.
`#2`—Collaboration ID
`#3`—Affiliation address
`#4`—Auxiliary info

```
\def\@affil@script#1#2#3#4{%
```

If the affiliation is the `\noaffiliation`, then we do nothing.

```
 \@ifnum{#1=\z@}{}{%
  \par
  \begingroup
   \frontmatter@affiliationfont
   \@ifnum{\c@affil<\affil@cutoff}{}{%
    \def\@thefnmark{#1}\@makefnmark
   }%
   \ignorespaces#3%
   \@if@empty{#4}{}{\frontmatter@footnote{#4}}%
   \par
  \endgroup
 }%
}%
\let\affil@cutoff\@ne
```

`\@author@present@script` This version of `\AU@opr` applies to the `superscriptaddress` class option. Need to add commas between groups of address numbers, which are passed in the macro `\@affilID@temp` in the form `{3}{4}{7}` if this set of authors is related to addresses, 3, 4 and 7.

```
\def\@author@present@script#1#2#3{%
 \begingroup
  \gdef\comma@space{\textsuperscript{,\,}}%
  \doauthor{#2}{#3}{\@affil@present@script}%
 \endgroup
 \advance\@tempcnta\m@ne
}%
```

```
\def\@affilcomma#1#2{%
 \@ifx{\z@#1}{%
  \@ifx{\relax#2}{}{%
   \@affilcomma{#2}%
  }%
 }{%
  #1%
  \@ifx{\relax#2}{}{%
   \@ifx{\z@#2}{%
    \@affilcomma
   }{%
    ,\,\@affilcomma{#2}%
   }%
  }%
 }%
}%
```

`\@affil@present@script`
`\@affil@present@script@`

The `\@affil@present@script` procedure presents the reference to the affiliations in the case of superscript addresses.

The affiliations are presented to `\@affil@present@script@`as a list of brace-delimited tokens, each containing the affiliation ID (a number). The list is terminated by the token `\relax`. The macro recurses (via tail recursion) until the terminating token is encountered. Note that an affiliation ID of zero signifies a `\noaffiliation`: references to the `\noaffiliation` are passed over.

Upon termination, the affiliations, if any, are produced via `\textsuperscript` and the appropriate boolean is queued via `\aftergroup`. The boolean signifies that a non-trivial affiliation was produced, so that the proper punctuation can be produced. The group in question is that opened and closed by the `\doauthor` procedure.

This procedure uses `\@tempa` as a scratch register.

```
\def\@affil@present@script{%
 \let\@tempa\@empty
 \expandafter\@affil@present@script@\@affilID@temp\relax
}%
\def\@affil@present@script@#1{%
 \@ifx{\relax#1}{%
  \@ifx{\@tempa\@empty}{%
   \aftergroup\false@sw
  }{%
   \textsuperscript{\expandafter\@affilcomma\@tempa\relax\relax}%
   \aftergroup\true@sw
  }%
 }{%
  \@ifnum{#1=\z@}{}{\appdef\@tempa{{#1}}}%
  \@affil@present@script@
 }%
}%
\@provide\@author@parskip{\z@skip}%
```

9.9.2 Authors and affiliations in groupedaddress style

\frontmatter@author@produce@group Alias for \frontmatter@author@producewhen groupedaddress is in effect, the procedure that formats the author/affiliation list.

In this case, the authors are being grouped above their shared addresses. How it works: the \@AFG@list, a list of all unique affiliation groups, is expanded. For each such group, all authors belonging to that group are formatted above the affiliation group. Collaborations are silently passed over.

This procedure must ensure that any paragraph that it starts gets ended finally.

```
\def\frontmatter@author@produce@group{%
  \begingroup
    \let\@author@present\@author@present@group
    \frontmatter@authorformat
    \frontmatterverbose@sw{\typeout{\string\frontmatter@author@produce@group:}\say\@AAC@list\sa
    \let\AU@temp\@empty
    \set@listcomma@list\AU@temp
    \def\CO@opr{\@collaboration@present\AU@temp}%
    \let\AFG@opr \affils@present@group
    \let\@listcomma\relax
    \@AFG@list
    \frontmatter@footnote@produce
    \par
  \endgroup
  \frontmatter@authorbelow
}%
\@provide\frontmatter@authorbelow{}%
```

\affils@present@group Alias of \AFG@opr, this procedure is used on every element of \@AFG@list, the list of unique affiliation groups, when formatting the author/affiliation block: It presents all authors sharing this affiliation group.

This procedure is called via the procedure \frontmatter@author@produce@group, an alias for \frontmatter@author@produce,

Its use pertains to the class options groupedaddress, unsortedaddress, and runinaddress (that is, \groupauthors@sw is true).

For each invocation, the entire \@AAC@list is run through (twice), making this process execute in N^2 time. (Is this really true?)

```
\def\affils@present@group#1{%
```

Count (using \@tempcnta) the authors in the author list whose affiliation group matches the argument #1.

Note that \AF@temp is used as a storage register for the argument of the procedure, for purposes of comparison using \ifx.

Note: here we use \@tempcnta to accumulate the number of authors in the current affiliation set.

QUERY: do the value of \AF@temp and \AF@opr need to persist?

QUERY: what value does \AU@opr have at this point? ANSWER: the \@affilID@count and \@affilID@match procedures assign meanings to \AU@opr

and \CO@opr dynamically. We initialize the two operators to \@undefined so as to catch the pathological cases.

```
\begingroup
  \def\AF@temp{#1}%
  \@tempcnta\z@
  \let\AU@opr \@undefined
  \let\CO@opr \@undefined
  \def\AF@opr{\@affilID@count\AF@temp\@tempcnta}%
  \@AAC@list
```

If there are no authors using this affiliation group, skip it.

```
  \@ifnum{\@tempcnta=\z@}{}{%
```

DPC: If we have a list of length two, need to locally zap a comma.

```
   \begingroup
    \frontmatter@above@affilgroup
    \set@listcomma@count\@tempcnta
```

Run through the author list, presenting (with \@author@present) those authors whose affiliation matched the given one.

```
    \let\AU@opr \@undefined
    \let\CO@opr \@undefined
    \def\AF@opr{\@affilID@match\AF@temp}%
    \@AAC@list
   \endgroup
```

Now that the authors have all been presented, present the affiliations, grouped.

Note: \@tempcnta is the number of affiliations for this set of authors.

```
   \begingroup
    \par
    \frontmatter@above@affiliation
    \frontmatter@affiliationfont
    \let\\\frontmatter@addressnewline
    \@tempcnta\z@
    \@tfor\AF@temp:=#1\do{%
     \expandafter\@ifx\expandafter{\expandafter\z@\AF@temp}{}{%
      \advance\@tempcnta\@ne
     }%
    }%
    \@ifnum{\@tempcnta=\tw@}{%
     \let\@listcomma\relax
    }{}%
    \def@after@address
    \runinaddress@sw{%
    }{%
     \tightenlines@sw{}{%
```

The following line determines the space between affilations in grouped mode

```
      \parskip\z@
     }%
     \appdef\after@address\par
```

```
      }%
      \let\AFF@opr \@affil@group
      \do@affil@fromgroup\@AFF@list#1\relax
    \endgroup
```

End of branch handling authors.

```
  }%
  \par
\endgroup
}%
```

`\def@after@address` `\def@after@address@empty`

Each of these procedures assigns a meaning to `\after@address`, the first a useful default, the second a nil. A society or journal substyle may define it according to its own requirements.

```
\def\def@after@address{\def\after@address{\@listcomma\ \@listand}}%
\def\def@after@address@empty{\let\after@address\@empty}%
```

`\@affilID@count` `\@affilID@match`

The control sequence name `\AF@opr` is placed in the author/affiliation list (`\@AAC@list`) for each (new) affiliation, with that affiliation as its argument.

The control sequence name `\AF@opr` is an alias of a procedure to count the number of authors at a given affiliation (`\@affilID@count`) or to present the given authors (`\@affilID@match`). These two procedures are executed when formatting under the `groupedaddress` class option.

```
\def\@affilID@count#1#2#3{%
  \def\@tempifx{#3}%
  \@ifx{\@tempifx#1}{%
    \def\AU@opr{\@author@count#2}%
  }{%
    \let\AU@opr \@author@gobble
  }%
  \let\CO@opr \@collaboration@gobble
}%
\def\@affilID@match#1#2{%
 \def\@tempifx{#2}%
 \@ifx{\@tempifx#1}{%
   \let\AU@opr \@author@present
 }{%
   \let\AU@opr \@author@gobble
 }%
  \let\CO@opr \@collaboration@gobble
}%
```

`\do@affil@fromgroup`

Executes iteratively: selects the next address ID to print, and then re-execute the list of addresses to print the text of that address.

Note: an argument of `\relax` breaks out of this iteration, while an argument of `\z@` (`\noaffiliation`) is a no-op.

Note: we have created an N^2 computing process. FIXME: can use hashing instead, you know.

```
\def\do@affil@fromgroup#1#2{%
  \@ifx{\relax#2}{}{%
    \count@#2\relax
    \@ifnum{\z@=\count@}{}{#1}%
    \do@affil@fromgroup#1%
  }%
}%
```

`\@affil@group` The alias of `\AFF@opr` when the list `\@AFF@list` is executed by `\do@affil@fromgroup` (`groupedaddress`): Print the address if its number matches `\count@` (value set by `\do@affil@fromgroup`).

Note that we detect a `\noaffiliation` here by the affiliation address being the same as the expansion of `\blankaffiliation`.

I made a mistake when coding the comparison to `\blankaffiliation`: it originally read `\@ifx{#3\blankaffiliation}`. This is obviously wrong once you think twice.

```
\def\@affil@group#1#2#3#4{%
  \@ifnum{#1=\count@}{%
   \def\@tempa{#3}%
   \@ifx{\@tempa\blankaffiliation}{}{%
    #3%
    \@if@empty{#4}{}{%
     \frontmatter@footnote{#4}%
    }%
    \after@address
   }%
   \advance\@tempcnta\m@ne
  }{}%
}%
```

`\@author@present@group` `\@affil@present@group` This version of `\AU@opr` applies to the `groupedaddress` class option and its ilk. Traverse the list calling the `\doauthor` hook on each author.

The third argument of `\doauthor` fulfills the requirement that it queue up a boolean via `\aftergroup`.

```
\def\@author@present@group#1#2#3{%
  \gdef\comma@space{\gdef\comma@space{\textsuperscript{,\,}}}%
  \doauthor{#2}{#3}{\@affil@present@group}%
  \advance\@tempcnta\m@ne
}%
\def\@affil@present@group{%
 \aftergroup\false@sw
}%
```

`\@pacs@produce` PACS, keywords and dates.

```
\def\@pacs@produce#1{%
 \showPACS@sw{%
  \begingroup
   \frontmatter@PACS@format
```

```
    \@pacs@name#1\par
   \endgroup
  }{%
   \@if@empty{#1}{}{%
    \class@warn{\PACS@warn}%
   }%
  }%
}%
\def\PACS@warn{If you want your PACS to appear in your output, use document class option showpa
```

\@keywords@produce

```
\def\@keywords@produce#1{%
 \showKEYS@sw{%
  \begingroup
   \frontmatter@keys@format
   \@keys@name#1\par
  \endgroup
 }{%
  \@if@empty{#1}{}{%
   \class@warn{If you want your keywords to appear in your output, use document class option sh
  }%
 }%
}%
```

atter@footnote@produce@footnote
matter@footnote@produce@endnote

Produce any footnotes to appear at the bottom of the title page.

If frontmatter footnotes have been redirected to the bibliography, this will be a no-op. ref.: \present@bibnote.

```
\def\frontmatter@footnote@produce@footnote{%
 \let\@TBN@opr\present@FM@footnote
 \@FMN@list
 \global\let\@FMN@list\@empty
}%

\def\present@FM@footnote#1#2{%
 \begingroup
  \csname c@\@mpfn\endcsname#1\relax
  \def\@thefnmark{\frontmatter@thefootnote}%
  \frontmatter@footnotetext{#2}%
 \endgroup
}%
\def\frontmatter@footnote@produce@endnote{%
}%
```

10 Package defaults

\title@column
\preprintsty@sw
\collaboration@sw
\frontmatter@footnote@produce
\do@output@MVL
\comma@space

We give meanings to the Booleans and other macros that a client document class would set on our behalf.

```
\appdef\frontmatter@init{%
```

```
\@ifxundefined\title@column         {\let\title@column\@empty}{}%
\@ifxundefined\preprintsty@sw       {\@booleanfalse\preprintsty@sw}{}%
\@ifxundefined\frontmatter@footnote@produce{\let\frontmatter@footnote@produce\frontmatter@foot
\@ifxundefined\do@output@MVL        {\let\do@output@MVL\@firstofone}{}%
\@ifxundefined\comma@space          {\let\comma@space\@empty}{}%
}%
```

`\frontmatter@thefootnote` In the title block, the footnote symbol (used for alternate affiliation and sundry) is a `\@fnsymbol` or a lowercase letter, as selected by a boolean. The society or journal may override this definition.

```
\def\frontmatter@thefootnote{%
 \altaffilletter@sw{\@alph}{\@fnsymbol}{\csname c@\@mpfn\endcsname}%
}%
\@ifx{\altaffilletter@sw\@undefined}{\@booleantrue\altaffilletter@sw}{}%
```

`\frontmatter@makefnmark` The default procedure for setting the footnote mark within the frontmatter. A client may override this procedure.

```
\def\frontmatter@makefnmark{%
 \@textsuperscript{%
  \normalfont\@thefnmark
 }%
}%
```

`\frontmatter@makefntext` This procedure is used when setting footnotes within the title block.

Note that we are `hyperref` friendly: we invoke `\Hy@raisedlink` so that this footnote becomes a hypertext anchor. If `hyperref` is not loaded, then `ltxutil` has our act covered via some stubs.

```
\long\def\frontmatter@makefntext#1{%
 \parindent 1em
 \noindent
 \Hy@raisedlink{\hyper@anchorstart{frontmatter.\expandafter\the\csname c@\@mpfn\endcsname}\hype
 \@makefnmark
 #1%
}%
```

`\frontmatter@setup` The default type specification for the title page. Clients will override this definition.

```
\def\frontmatter@setup{}%
```

`\frontmatter@RRAPformat` `\punct@RRAP` `\produce@RRAP` The procedure `\frontmatter@RRAPformat` is the formatting procedure for the "Received, Revised, etc." block on the title page. The macro `\punct@RRAP` produces the punctuation between the entries. Clients will override these two definitions.

The procedure `\produce@RRAP` is fairly general, and should be used by most journal substyles.

The procedure `\frontmatter@RRAP@format` will be overridden by the journal.

```
\def\frontmatter@RRAPformat#1{%
 \removelastskip
```

```
 \begingroup
  \frontmatter@RRAP@format
  #1\par
 \endgroup
}%
\def\punct@RRAP{; }%
\def\produce@RRAP#1{%
  \@if@empty{#1}{}{%
   \@ifvmode{\leavevmode}{\unskip\punct@RRAP\ignorespaces}%
   #1%
  }%
}%
```

`\frontmatter@authorformat` The default type specification for the author list. Clients will override this definition. For one alternative, see the `revtex4` document class.

```
\def\frontmatter@authorformat{}%
```

`\frontmatter@above@affilgroup` The default amount of space above an affiliation group (in a `\affils@present@group` production). Clients will override this definition.

```
\def\frontmatter@above@affilgroup{}%
```

`\frontmatter@above@affiliation` `matter@above@affiliation@script` The `\frontmatter@above@affiliation` is the amount of space above affiliations in the `groupedaddress` style, and `\frontmatter@above@affiliation@script` is that for `superscriptaddress`. Clients will override this definition. For one alternative, see the `revtex4` document class.

```
\def\frontmatter@above@affiliation{}%
\def\frontmatter@above@affiliation@script{}%
```

`\frontmatter@affiliationfont` The default type specification for the affiliation. Clients will override this definition. For one alternative, see the `revtex4` document class, where several alternative definitions are made.

```
\def\frontmatter@affiliationfont{\itshape\selectfont}%
```

`\frontmatter@RRAP@format` The default type specification for the dates. Clients will override this definition.

```
\def\frontmatter@RRAP@format{}%
```

`\frontmatter@PACS@format` Specify the formatting of the title page PACS statement. Journals will override.

```
\def\frontmatter@PACS@format{}%
```

`\frontmatter@keys@format` Specify the formatting of the title page keywords statement. Journals will override.

```
\def\frontmatter@keys@format{}%
```

`\frontmatter@finalspace`

```
\def\frontmatter@finalspace{\addvspace{18\p@}}
```

`\frontmatter@addressnewline` The definition of \\ for address handling. Default puts all the 'lines' on a run-in line, separated by comma and space. DPC: was space between lines of addresss.

```
\def\frontmatter@addressnewline{%
  \@ifhmode{\skip@\lastskip\unskip\unpenalty\break\hskip\skip@}{}%
  % was: \vskip-.5ex
}%
```

`\frontmatter@preabstractspace` `\frontmatter@postabstractspace` Space above and space below abstract in title block

```
\def\frontmatter@preabstractspace{5.5\p@}
\def\frontmatter@postabstractspace{6.5\p@}
```

`\aftermaketitle@chk` Error-checking code.

```
\def\aftermaketitle@chk#1{%
  \@ifx{\maketitle\relax}{%
    \class@err{\protect#1 must be used before \protect\maketitle}%
  }{}%
}%
```

`\ps@titlepage` Default page style for title page. Journals will override this procedure.

```
\def\ps@titlepage{\ps@empty}%
```

FIXME: there is a limitation to the default meaning of `\@startpage`: the label "FirstPage" is only defined if the document has a `\maketitle` command.

```
\def\volumeyear#1{\gdef\@volumeyear{#1}}%
\def\@volumeyear{}%
\def\volumenumber#1{\gdef\@volumenumber{#1}}%
\def\@volumenumber{}%
\def\issuenumber#1{\gdef\@issuenumber{#1}}%
\def\@issuenumber{}%
\def\eid#1{\gdef\@eid{#1}}%
\def\@eid{}%
%
\def\startpage#1{\gdef\@startpage{#1}\c@page#1\relax}%
\def\@startpage{\pageref{FirstPage}}%
\def\endpage#1{\gdef\@endpage{#1}}%
\def\@endpage{\pageref{LastPage}}%
```

10.1 Printing out the "list-of" elements

FIXME: The `\appendix@toc` procedure should change the meaning of `\l@section` so that the `\section`s can be appropriately formatted, reflecting their status as appendices.

```
\def\print@toc#1{%
 \begingroup
  \expandafter\section
  \expandafter*%
  \expandafter{%
```

```
              \csname#1name\endcsname
              }%
  \let\appendix\appendix@toc
  \@starttoc{#1}%
 \endgroup
}%
\def\appendix@toc{}%
```

\Dated@name
\Received@name
\Revised@name
\Accepted@name
\Published@name

These strings are used in the \date, et al. commands.

```
\def\Dated@name{Dated }%
\def\Received@name{Received }%
\def\Revised@name{Revised }%
\def\Accepted@name{Accepted }%
\def\Published@name{Published }%
```

Two commands require robustifying against harsh treatment when TOC entries are written out. Note that \class@warn is not user-level markup, but crops up when the \title command is missing from the document.

```
\appdef\robustify@contents{%
 \let\thanks\@gobble\let\class@warn\@gobble
 \def\begin{\string\begin}\def\end{\string\end}%
}%
```

10.2 Syntax switch

Shall we take over the syntax of LaTeX? By default, Yes, unless the Boolean below has already been defined. In the package version, this is controlled by an option; in the kernel version, the Boolean remains undefined.

\maketitle presents a complication. If this code will execute under ltxgrid, then its version of the procedure will do fine.

As a package, under article, things are different. In this case, what should be done depends upon \if@titlepage. If true, then just execute \maketitle as it stands. If false, then the article class has defined its wrapper procedure, and we should take over the meaning of \@maketitle instead.

Here is a list of the commands defined by this package that potentially override those of standard LaTeX: \frontmatter@title, \frontmatter@author, \frontmatter@and, \frontmatter@thanks, \frontmatter@date, frontmatter@abstract, frontmatter@titlepage, \frontmatter@maketitle.

The following code will perform that override, given the appropriate state of the Boolean.

```
\@ifxundefined\frontmatter@syntax@sw{\@booleantrue\frontmatter@syntax@sw}{}%
\frontmatter@syntax@sw{%
 \let\title         \frontmatter@title
 \let\author        \frontmatter@author
 \let\date          \frontmatter@date
 \@ifxundefined\@maketitle{%
  \let\maketitle    \frontmatter@maketitle
  \@booleantrue     \titlepage@sw
```

```
}{%
 \let\@maketitle   \frontmatter@maketitle
 \prepdef\maketitle\@author@finish
}%
\let\noaffiliation \frontmatter@noaffiliation
\let\thanks@latex  \thanks
\let\thanks        \frontmatter@thanks
\let\and@latex     \and
\let\and           \frontmatter@and
\let@environment{titlepage}{frontmatter@titlepage}%
\let@environment{abstract}{frontmatter@abstract}%
}{%
\let\noaffiliation\@empty
}%
```

10.3 End of the `ltxfront` kernel

The kernel portion of the `ltxfront` package is complete.

```
%</kernel>
```

10.4 Remainder of the `ltxfront` package

Include any code here that ought to be incorporated into the package, but should not be part of the kernel.

```
%<*package>
```

Define formatting as like as possible to the article class. Comands that prepare for `\maketitle`: `\author`, `\title`, `\date`.

Argument of the above three may contain `\\` and `\thanks`; argument of `\author`may contain `\and`. The `\\` may still work fine; `\and` will be ignored (warning); `\thanks` even though it should not reside in the argument, will work.

An explicit `\footnote`should be treated as if it were `\thanks`.

```
% \@maketitle: \frontmatter@footnote@produce
%
```

```
\def\thanks@latex#1{%
 \footnotemark
 \expandafter\expandafter
 \expandafter\gappdef
 \expandafter\expandafter
 \expandafter\@thanks
 \expandafter\expandafter
 \expandafter{%
 \expandafter\expandafter
 \expandafter\footnotetext
 \expandafter\expandafter
 \expandafter[%
```

```
 \expandafter\the\csname c@\@mpfn\endcsname]{#1}}%
}%
\@booleanfalse\altaffilletter@sw
\@if@sw\if@titlepage\fi{\@booleantrue}{\@booleanfalse}\titlepage@sw
\def\frontmatter@title@above{\newpage\null\vskip2em\relax}%
\def\frontmatter@title@format{\centering\LARGE\let\thanks\thanks@latex}%
\def\frontmatter@title@below{\vskip1.5em\relax}%
\def\frontmatter@authorformat{\centering\large\advance\baselineskip\p@\parskip11.5\p@\let\thank
\def\frontmatter@authorbelow{\vskip 1em\relax}%
\def\frontmatter@above@affiliation{}%
\def\frontmatter@above@affiliation@script{}%
\def\frontmatter@affiliationfont{\centering\itshape}%
\def\frontmatter@RRAP@format{\centering\large}%
\def\frontmatter@preabstractspace{1.5em}%

\long\def\frontmatter@footnotetext{%
 \expandafter\expandafter
 \expandafter\footnotetext
 \expandafter\expandafter
 \expandafter[%
 \expandafter\the\csname c@\@mpfn\endcsname]%
}%
\def\and@space{\\}%
\def\andname{and}%
```

Implement the feature of the article class whereby there is a default `\@date`, left over from the LaTeX kernel. This package does not have a default date. The user must put in an explicit command, like `\date{`⟨*\today*⟩`}`, which will retore LaTeX's default behavior.

What if the abstract environment falls after the `\maketitle` command? The environment will be undefined; the user will have to deal with this.

End of the package.

```
%</package>
```

Here ends the programmer's documentation.

Index

Symbols

\% 556
\, ... 380, 1000, 1016, 1143
.dtx 4
\@AAC@list . 26–28, 31, 42, 47, 49
\@AAC@list 502, 503, 508–510, 527, 536, 538, 833, 921, 927, 935, 1043, 1063, 1071
\@AF@join 29
\@AF@join 276, 316, 392, 546, 563, 568, 574, 577, 836
\@AF@join@error . . 547, 549, 836
\@AFF@list . 24, 25, 28, 42, 44, 50
\@AFF@list . . 428, 453, 538, 834, 921, 942, 1043, 1096
\@AFG@list 25–28, 44, 47
\@AFG@list 504, 521, 523, 524, 528, 536, 538, 835, 921, 1043, 1049
\@AFU@list 44
\@FMN@list 32, 33
\@FMN@list . 634, 651, 654, 1177, 1178
\@TBN@opr 32
\@TBN@opr 633, 651, 1176
\@accepted 676, 885
\@address 838
\@affil 23–25, 30
\@affil . . 408, 411, 413, 415, 418, 420, 426, 431, 442, 452
\@affil@addr@def 426, 463
\@affil@aux@def 431, 466
\@affil@cleared 402
\@affil@def 23
\@affil@def 393, 402
\@affil@group 28
\@affil@group 1095, 1129
\@affil@init 23, 24
\@affil@init 402, 460
\@affil@join 392, 402
\@affil@join@ 402
\@affil@match 24, 25, 28
\@affil@match 427, 473
\@affil@nil 24
\@affil@nil 402, 442
\@affil@none 367, 398, 402
\@affil@present@group 32
\@affil@present@group . . . 1142
\@affil@present@script . 31, 46
\@affil@present@script . . 1001, 1021
\@affil@present@script@ . . . 46
\@affil@present@script@ . 1021
\@affil@script 31
\@affil@script 941, 983
\@affilID@count 47, 49
\@affilID@count 1062, 1104
\@affilID@def 43
\@affilID@def 931, 956, 968
\@affilID@match 47, 49
\@affilID@match 1070, 1104
\@affilID@temp 43–45
\@affilID@temp . . 968, 969, 1023
\@affilclear@sw 23
\@affilclear@sw 402, 422
\@affilcomma . 1005, 1008, 1014, 1016, 1030
\@affiliation 388, 390
\@affils@sw 16, 17, 24
\@affils@sw . 239, 244, 250, 256, 262, 425
\@argswap 328, 330
\@argswap@val 333, 335, 444, 448, 451, 452, 702, 906
\@author 20, 23, 30
\@author . 306, 309, 318, 321, 323, 334
\@author@cleared 20
\@author@cleared 303, 928
\@author@count . . 925, 948, 1107
\@author@def 295, 298, 303
\@author@finish 23
\@author@finish . 344, 827, 1292
\@author@gobble 20
\@author@gobble . . 304, 520, 950, 966, 1109, 1118

\@author@init 20
\@author@init 303, 338
\@author@join 303
\@author@join@ 303
\@author@parskip 1038
\@author@present 16, 30, 48
\@author@present . 920, 932, 958, 1041, 1116
\@author@present@group 32
\@author@present@group . . 1041, 1142
\@author@present@script . . . 31
\@author@present@script . . 920, 998
\@authorclear@sw 20
\@authorclear@sw 303, 325
\@booleanfalse 235, 236, 238, 240, 245, 250, 251, 261, 263, 301, 307, 423, 518, 630, 1191, 1320, 1321
\@booleantrue 234, 237, 239, 243, 244, 249, 255–257, 262, 298, 490, 613, 622, 1199, 1282, 1289, 1321
\@collaboration@count . 926, 964
\@collaboration@gobble . . . 601, 1111, 1120
\@collaboration@present . . 933, 952, 1046
\@currentlabel 736
\@date 57
\@date 676, 882
\@eid 1254, 1255
\@email 560, 561
\@empty 28
\@endpage 1259, 1260
\@finalstrut 739
\@firstofone . . . 377, 378, 1193
\@fnsymbol 52
\@fnsymbol 1197
\@footnotemark 32
\@footnotemark 848, 850
\@footnotetext . . . 712, 849, 851
\@homepage 30
\@homepage 565, 566
\@if@empty . . 374, 606, 972, 977, 992, 1134, 1157, 1170, 1222
\@if@sw 1321
\@ifdim 801
\@ifhmode 1237
\@ifnextchar 852
\@ifnum . . 478, 587, 596, 598–600, 984, 988, 1034, 1064, 1084, 1125, 1130
\@ifundefined 29
\@ifvmode 1223
\@ifx 50
\@ifx 309, 347, 348, 353, 359, 413, 432, 433, 442, 476, 499, 500, 511, 512, 526, 612, 621, 808, 860, 907, 971, 1006, 1007, 1012, 1013, 1026, 1027, 1080, 1106, 1115, 1123, 1132, 1199, 1243
\@ifx@empty 889, 892
\@ifxundefined 1190–1194, 1282, 1287
\@iiiparbox 35
\@issuenumber 1252, 1253
\@join 22
\@join 285, 321, 373, 418
\@keys@name 1167
\@keywords 669, 892, 893
\@keywords@produce . . 893, 1163
\@listand 30
\@listand 598, 607, 1102
\@listcomma 30, 31, 42, 43
\@listcomma . 581, 583, 588, 590, 599, 603, 1048, 1085, 1102
\@listcomma@comma 583, 590, 599
\@listcomma@comma@UK 600
\@listdepth 715
\@makefnmark 663, 872, 975, 989, 1209
\@makefntext 36
\@makeother 556–558
\@maketitle 55
\@maketitle 1287, 1291
\@minipagefalse 727
\@minipagerestore 717
\@mpfn 639, 644, 661, 711,

736, 869, 1182, 1197, 1208, 1318, 1337
\@mpfootnotetext 36
\@mplistdepth 715
\@ne 31, 45
\@pacs 686, 889, 890
\@pacs@name 1154
\@pacs@produce 890, 1150
\@parboxrestore 710, 735
\@preprint 693, 695
\@provide 1038, 1055
\@published 676, 886
\@received 34
\@received 676, 883
\@reset@ac 580, 593
\@revised 676, 884
\@secondofthree . . 356, 357, 372
\@secondoftwo 355
\@separator . . 285, 321, 376, 418
\@setminipage 718
\@shorttitle 275, 278
\@startpage 54
\@startpage 1257, 1258
\@starttoc 1269
\@tempa 24, 25, 28, 42–44, 46
\@tempa 426, 427, 435, 631, 633, 971, 1022, 1027, 1030, 1034, 1131, 1132
\@tempb 25, 32, 33
\@tempb . . 431–433, 623, 632, 637
\@tempc 24, 25
\@tempc 443, 447, 458, 491
\@tempcnt 42
\@tempcnta . . . 30, 31, 42, 47, 48
\@tempd 24, 25
\@tempd 424, 433, 492
\@tempifx 475, 476, 611, 612, 620, 621, 971, 1105, 1106, 1114, 1115
\@textsuperscript 1201
\@tfor 1079
\@thanks 1311
\@thefnmark . 662, 736, 975, 989, 1183, 1202
\@title 274, 278, 904, 906
\@title@aux . 281, 283, 285, 907, 908
\@title@aux@cleared . 281, 283, 907
\@title@join 276, 283
\@title@join@ 283
\@undefined 48
\@undefined 841, 1060, 1061, 1068, 1069, 1199
\@volumenumber 1250, 1251
\@volumeyear 1248, 1249
\@x@sf 658, 666
\\ 54, 56
\{ 137
\cmd␣␣October␣6,␣2020, argument 57
\} 137
_ 50, 558
\~ 557
00readme 4
10pt document class option . . . 69
11pt document class option . . . 69
12pt document class option . . . 69

␣ 69, 72
\␣ . 38, 44, 67, 68, 72, 91, 97, 103, 107, 116, 144, 216, 1102

A

\absbox 35
\absbox 697, 706, 799, 805
\abstract 841
abstract (environment) 698
abstract environment . 2, 14, 35
\abstractname 702, 764
\accepted 34
\accepted 676
\Accepted@name 682, 1273
\active 380
\active@comma 381, 382
\add@AUCO@grp 21
\add@AUCO@grp 324
\add@list@val@val@val 25
\add@list@val@val@val . 453, 469
\addcontents@abstract . 699, 722

\addcontentsline 702, 906
⟨*address*⟩ placeholder 29
\AF@grp 25–28
\AF@grp . . 353, 458, 459, 498, 499, 509, 519, 524, 534, 538
\AF@opr 26–28, 43, 44, 47, 49
\AF@opr . 502, 508, 924, 931, 1062, 1070
\AF@temp 47
\AF@temp . 1058, 1062, 1070, 1079, 1080
\AFF@opr . . . 23, 25, 28, 44, 45, 50
\AFF@opr 427, 453, 941, 1095
\affil@cutoff 31, 45
\affil@cutoff 988, 997
\affil@present@script 970
\affil@script 28, 44
\affil@script 970
\affiliation . 10, 12, 13, 22, 23, 28, 29
affiliation, argument 12
\affiliation . 204, 208, 380, 831
⟨*affiliation ID*⟩ placeholder . . . 29
\affils@present@group . . 31, 53
\affils@present@group . . . 1047, 1056
\affils@present@script . . . 970
\AFG@opr 27, 28, 32, 44, 47
\AFG@opr 519, 523, 1047
\after@address 49
\after@address 1093, 1102, 1103, 1137
\aftergroup 31, 46, 50
\aftergroup . . . 1028, 1031, 1148
\aftermaketitle@chk . 670, 687, 705, 1242
\altaffiliation 12, 13, 18, 29, 31
\altaffiliation 551, 576
\altaffilletter@sw 1197, 1199, 1320
\and 19, 22, 56
\and . . 379, 830, 1297, 1298, 1325
\and@latex 1297
\and@space 1325, 1339
\andname 598, 1340
\appdef 21, 25
\appdef 271, 278, 290, 300, 310, 341, 404, 470, 502, 508, 523, 538, 546, 570, 653, 673, 690, 694, 811, 859, 914, 1034, 1093, 1189, 1278
\appdef@e 503, 504, 510, 527, 528
\appdef@eval . 458, 509, 524, 965
\appendix 1268
\appendix@toc 54
\appendix@toc 1268, 1272
argument
    \cmd␣␣October␣6,␣2020 . 57
    affiliation 12
    e-mail␣address 12
    miscellaneous␣text 12
    title 10
    URL 12
argument, optional
    optional␣text 12
article document class 19, 38, 55
\AU@grp 20, 21, 26–28
\AU@grp . . 330, 347, 361, 498, 510, 511, 533, 538
\AU@opr 21, 28, 43, 45, 47, 50
\AU@opr 330, 356, 925, 932, 1060, 1068, 1107, 1109, 1116, 1118
\AU@temp 42
\AU@temp 922, 926, 933, 934, 1044–1046
auguide.tex 3
\author . 10–13, 19, 22, 27–29, 31, 56
⟨*author*⟩ placeholder 29
\author . . 48, 202, 206, 294, 314, 832, 1285
⟨*auxiliary info*⟩ placeholder . . . 29
⟨*auxilliary info*⟩ placeholder . . 29

B

\baselineskip 798, 814, 815, 917, 1325
bibnotes document class option 14
\blankaffiliation 50
\blankaffiliation . . . 401, 1132
\break 1237

C

\c@affil 19
\c@affil 288, 446–448, 988
\c@collab 19
\c@collab 288, 327, 335, 451, 478
\c@mpfootnote 711, 871
\c@page 1257
\cat@comma@active 380, 381, 387
\catcode 380
\centering 763, 1323, 1325, 1329, 1330
\changes 141–192
\class@documenthook 18
\class@documenthook . 271, 570, 859
\class@err 379, 1244
\class@info 861
\class@name 230
\class@warn 55
\class@warn . 279, 360, 363, 434, 550, 756, 1158, 1171, 1279
⟨*class customization commands*⟩ placeholder 10
\classname . 59, 66, 125, 127, 129
\classoption 188
\clearpage 844
\clo@groupedaddress . . 241, 242
\clo@runinaddress 253, 254
\clo@superscriptaddress . . 259, 260
\clo@unsortedaddress . 247, 248
\cmd 137
\CO@grp 20, 26–28, 31
\CO@grp . . 328, 348, 359, 364, 498, 500, 503–505, 512, 526–529, 532, 540
\CO@opr 27, 28, 48
\CO@opr 328, 357, 520, 926, 928, 933, 1046, 1061, 1069, 1111, 1120
\collaboration 10, 12, 13, 19, 20, 24, 26–29
\collaboration 297, 314
\collaboration@sw 298, 301, 307, 326, 1189
⟨*collaboration ID*⟩ placeholder . 29
\color@begingroup 707, 737
\color@endgroup 728, 741
\columnwidth 708, 709
\comma@space . . 606, 1000, 1143, 1189
\copyright 44
\count 43
\count@ 50
\count@ 1124, 1125, 1130
\cs . 141, 144, 146, 147, 149, 153, 161, 162, 165, 166, 168–170, 172, 173, 176–181, 184–190
\csname 29, 33, 39, 43
\csname 30, 644, 661, 701, 736, 1182, 1197, 1208, 1266, 1318, 1337

D

\date 10, 34, 55–57
⟨*date*⟩ placeholder 10
\date 210, 676, 1286
\Dated@name 676, 1273
\def@after@address . 1087, 1102
\def@after@address@empty 1102
\dimen@ 798, 800, 801
\do 1079
\do@affil@fromgroup 50
\do@affil@fromgroup 1096, 1122
\do@output@MVL 788, 1189
\do@title 273
\doauthor 31, 32, 44, 46, 50
\doauthor 602, 1001, 1144
doc 4
\DocInput 9
docstrip document class 16
document class
  article 19, 38, 55
  docstrip 16
  frontmatter 18
  hyperref 32, 35, 36, 39, 52, 74
  lineno 73, 75
  ltxdoc 4, 9
  ltxfront 2, 15, 56
  ltxgrid 55, 75
  ltxutil 16, 36, 52
  multicol 69

revtex4 53
revtex4-1 1
document class option
10pt 69
11pt 69
12pt 69
bibnotes 14
groupauthors 69
groupedaddress . 11, 17, 28, 31, 47, 49, 50, 53
hypertext 73
nobibnotes 14
noshowkeys 15
noshowpacs 14
preprint 14
runinaddress 16, 17, 47
showkeys 15
showpacs 14
superscriptaddress . 11, 12, 17, 22, 25, 31, 41, 43–45, 53, 74
unsortedaddress 11, 12, 16, 17, 47
document environment 4
\documentclass 15

E

e-mail␣address, argument . . . 12
\edef 658
\eid 1254
\email 10, 12, 13, 18, 29, 31
\email 551, 555, 837
\endabstract 35
\endabstract 841
\endcsname 39
\endcsname 30, 644, 661, 701, 736, 1182, 1197, 1208, 1266, 1318, 1337
\endminipage 35
\endpage 1259
environment
abstract 2, 14, 35
document 4
frontmatter@abstract . . 55
frontmatter@titlepage . 55
minipage 35
thebibliography 75
environments:
abstract 698
titlepage 816
\everypar 35

F

\false@sw 17, 31, 32
\false@sw . . . 350, 481, 487, 514, 1028, 1148
file
.dtx 4
00readme 4
␣ 69, 72
auguide.tex 3
doc 4
ltxfront 2, 56
package 2, 15
README 3
template.aps 3
\file . 77, 80, 86, 87, 96, 103, 104, 107, 110, 111, 114, 115, 118, 119, 125, 127, 129, 131, 134
\firstname 14
\firstname 377
\footnote 56
\footnote 358, 721, 749, 750
\footnotemark . . . 751, 752, 1307
\footnotesep 739
\footnotetext . . . 753, 754, 1315, 1334
frontmatter document class . . 18
\frontmatter@above@affilgroup 1066, 1228
\frontmatter@above@affiliation 53
\frontmatter@above@affiliation . . . 1075, 1229, 1327
\frontmatter@above@affiliation@script 53
\frontmatter@above@affiliation@script 940, 1229, 1328
frontmatter@abstract environment 55
\frontmatter@abstract@produce 785, 888

\frontmatter@abstractfont 35, 36
\frontmatter@abstractfont 720, 768
\frontmatter@abstractheading 35
\frontmatter@abstractheading 719, 761
\frontmatter@abstractwidth 35
\frontmatter@abstractwidth 784
\frontmatter@addressnewline 1077, 1236
\frontmatter@affiliationfont 45
\frontmatter@affiliationfont . 987, 1076, 1231, 1329
\frontmatter@and 55
\frontmatter@and 379, 1298
\frontmatter@author 55
\frontmatter@author . 294, 1285
\frontmatter@author@produce 41, 47
\frontmatter@author@produce@group 47
\frontmatter@author@produce@group 877, 1039
\frontmatter@author@produce@script 41, 43
\frontmatter@author@produce@script 879, 918
\frontmatter@authorbelow 1053, 1055, 1326
\frontmatter@authorformat 930, 1042, 1227, 1325
\frontmatter@collaboration@above 955, 963
\frontmatter@date 55
\frontmatter@date . . . 676, 1286
\frontmatter@description . 769
\frontmatter@descriptionlabel 774, 779
\frontmatter@finalspace . . 896, 1235
\frontmatter@foot@mark 33
\frontmatter@foot@mark 641, 649
\frontmatter@footnote 606, 628, 908, 977, 992, 1135
\frontmatter@footnote@produce 943, 1050, 1189
\frontmatter@footnote@produce@endnote 1175
\frontmatter@footnote@produce@footnote 1175, 1192
\frontmatter@footnotemark 636, 650, 656
\frontmatter@footnotetext 712, 731, 1184, 1332
\frontmatter@init 18
\frontmatter@init 271, 278, 290, 300, 310, 404, 538, 546, 653, 673, 690, 694, 811, 1189
\frontmatter@keys@format 1166, 1234
\frontmatter@makefnmark . . 872, 1200
\frontmatter@makefntext . . . 36
\frontmatter@makefntext . . 738, 1205
\frontmatter@maketitle 55
\frontmatter@maketitle . . . 826, 862, 1288, 1291
\frontmatter@noaffiliation 395, 1294
\frontmatter@PACS@format 1153, 1233
\frontmatter@postabstractspace 809, 814, 1240
\frontmatter@preabstractspace 789, 795, 814, 1240, 1331
\frontmatter@RRAP@format . . 52
\frontmatter@RRAP@format 1216, 1232, 1330
\frontmatter@RRAPformat . . . 52
\frontmatter@RRAPformat . . 881, 1213
\frontmatter@setup . . 873, 1212
\frontmatter@syntax@sw . . . 236, 237, 1282, 1283
\frontmatter@thanks 55
\frontmatter@thanks . 573, 1296
\frontmatter@thefootnote . 662, 1183, 1196

\frontmatter@title 55
\frontmatter@title . . 273, 1284
\frontmatter@title@above . . 41
\frontmatter@title@above . 902, 915, 1322
\frontmatter@title@below . . 41
\frontmatter@title@below . 911, 915, 1324
\frontmatter@title@format . 41
\frontmatter@title@format 903, 915, 1323
\frontmatter@title@produce 875, 899
frontmatter@titlepage environment 55
\frontmatterverbose@sw 16
\frontmatterverbose@sw . . . 234, 235, 314, 342, 345, 361, 364, 384, 396, 459, 471, 498, 536, 921, 1043

G

\g@addto@macro 651
\gappdef 693, 1309
\GetFileInfo 35
groupauthors document class option 69
\groupauthors@sw . 16, 17, 25, 47
\groupauthors@sw . 240, 243, 249, 255, 261, 477, 876
\groupedaddress 241
groupedaddress document class option 11, 17, 28, 31, 47, 49, 50, 53

H

\H@@footnotemark 850
\H@@footnotetext 851
\homepage . . 10, 12, 13, 18, 29–31
\homepage 551, 565
\href 355, 563, 568, 570
\ht 800
\Hy@maketitle@end 856
\Hy@maketitle@optarg 853
\Hy@raisedlink 52
\Hy@raisedlink 1208
\Hy@saved@footnotemark . . . 848
\Hy@saved@footnotetext . . . 849
\HyOrg@maketitle 855
\hyper@anchorend 1208
\hyper@anchorstart 1208
\hyper@linkend 664
\hyper@linkstart 660
hyperref document class . 32, 35, 36, 39, 52, 74
hypertext document class option 73

I

\if@titlepage 55
\if@titlepage 1321
\iffalse 52
\ifx 47
\ignorespaces 603, 739, 958, 975, 991, 1223
\issuenumber 1252
\item 101, 106, 117, 120, 124, 126, 128, 133, 135
\itemindent 773
\itshape 1231, 1329

K

\keywords 14, 34
\keywords 669

L

\l@section 54
\label 874
\labelsep 780
\labelwidth 772
\LARGE 1323
\large 763, 1325, 1330
\lastskip 1237
\LaTeX . 38, 67, 68, 78, 88, 91, 107, 113, 116, 118, 144, 216
\LaTeXe 72, 97, 227
\leftmargin 771
\let@environment 716, 1299, 1300
\let@mark 914
lineno document class . . . 73, 75
\listand 42
\listand 598
\listcomma 42

\ltx@footnote@pop 868
\ltx@no@footnote 744
\ltx@no@footwarn 744
\ltx@no@xfootmark 746, 751
\ltx@no@xfootnote 745, 749
\ltx@no@xfoottext 747, 753
\ltx@no@yfootmark 746, 752
\ltx@no@yfootnote 745, 750
\ltx@no@yfoottext 747, 754
\ltx@xfootmark 746
\ltx@xfootnote 745
\ltx@xfoottext 747
\ltx@yfootmark 746
\ltx@yfootnote 745
\ltx@yfoottext 747
ltxdoc document class 4, 9
ltxfront 2, 56
ltxfront document class 2, 15, 56
ltxgrid document class . . 55, 75
ltxutil document class 16, 36, 52

M

\makeatletter 28
\makeatother 31
\makelabel 774
\maketitle 10, 14, 15, 18, 34, 35, 39, 54–57
\maketitle 57, 213, 826, 860–862, 1243, 1244, 1288, 1292
\maketitle@Hy 847
\mini@note 36
\mini@note 721
\mini@notes 808, 811
\minipage 35
minipage environment 35
\minipagefootnote@drop . . . 742
\minipagefootnote@here . . . 726
\minipagefootnote@init . . . 713
\minipagefootnote@pick . . . 732
miscellaneous␣text, argument 12
\move@AF 19, 25
\move@AF . 315, 346, 367, 385, 397, 398, 421
\move@AU 19–21
\move@AU . 315, 324, 346, 385, 397
\move@AUAF 19
\move@AUAF . . . 315, 370, 399, 497
\mp@footnotetext 36
multicol document class 69

N

\newbox 697
\newcounter 19
\noaffiliation 12, 21, 23, 24, 26, 45, 46, 49, 50
\noaffiliation . . . 360, 363, 395, 1294, 1302
nobibnotes document class option 14
\normalfont 781, 1202
noshowkeys document class option 15
noshowpacs document class option 14

O

\onecolumngrid 817
optional␣text, optional argument 12
\OptionNotUsed 267

P

package 2, 15
\package@name 225, 226, 230
\PackageInfo 226
\pacs 14
\pacs 686
\PACS@warn 1158, 1162
\pageref 1258, 1260
\parindent 35
\parindent 1206
\parskip 35
\parskip 1091, 1325
\phantomsection 700, 906
placeholder
⟨*address*⟩ 29
⟨*affiliation ID*⟩ 29
⟨*author*⟩ 29
⟨*auxiliary info*⟩ 29
⟨*auxilliary info*⟩ 29
⟨*class customization commands*⟩ 10

⟨*collaboration ID*⟩ 29
⟨*date*⟩ 10
⟨*your document here*⟩ 9
\post@absbox 806, 813
\prep@absbox 804, 812
\prepdef 1292
\preprint 15
\preprint 693
preprint document class option 14
\preprintsty@sw 787, 1189
\present@bibnote 51
\present@FM@footnote 1176, 1180
\prevdepth 800
\print@toc 1261
\produce@RRAP 52
\produce@RRAP . . . 882–886, 1213
\protected@edef 736
\providecommand 571
\ProvidesFile 6
\ps@empty 1247
\ps@titlepage 1247
\published 34
\published 676
\Published@name 684, 1273
\punct@RRAP 52
\punct@RRAP 1213

R

README 3
\received 34
\received 676
\Received@name 678, 1273
\RecordChanges 33
\relax 23, 39, 46, 49
\removelastskip 1214
\RequirePackage . 25, 27, 30, 231
\revised 34
\revised 211, 676
\Revised@name 680, 1273
revtex4 document class 53
revtex4-1 document class 1
\robustify@contents 1278
\rule 739
\runinaddress 253
runinaddress document class option 16, 17, 47
\runinaddress@sw 16, 17
\runinaddress@sw . 238, 245, 251, 257, 263, 1088

S

\sanitize@url 555, 560, 565
\say 342, 361, 364, 459, 471, 498, 536, 921, 1043
\sc 88
\section 54
\section 84, 215, 1263
\set@footnotefont 733
\set@footnotewidth 734
\set@listcomma@count 30
\set@listcomma@count 579, 1067
\set@listcomma@list 30
\set@listcomma@list . 579, 934, 961, 1045
\set@listindent 714
\set@listindent@ 714
showkeys document class option 15
\showKEYS@sw 1164
showpacs document class option 14
\showPACS@sw 1151
\skip@ 1237
\spacefactor 658, 666
\special 35
\startpage 1257
\stepcounter 639
\StopEventually 5
\string . . 314, 345, 360, 363, 384, 396, 551, 757, 758, 861, 921, 1043, 1280
\strutbox 739
\subsection 94, 140
\superscriptaddress 259
superscriptaddress document class option 11, 12, 17, 22, 25, 31, 41, 43–45, 53, 74
\suppressfloats 829
\surname 14
\surname 377

T

\tableofcontents 82

\temp@sw 24, 25, 33
\temp@sw . 423, 429, 474, 490, 518, 522, 610, 613, 619, 622, 630, 635
template.aps 3
\TeX 72, 103, 118
\textsuperscript 46
\textsuperscript . . 1000, 1030, 1143
\texttt . . 50, 114, 115, 122, 131, 134, 137, 164, 165, 184
\textwidth 708, 784
\thanks . 10, 12, 13, 18, 29–31, 56
\thanks 39, 43, 50, 203, 207, 551, 573, 758, 840, 1279, 1295, 1296, 1323, 1325
\thanks@latex . 1295, 1306, 1323, 1325
\the 19, 33
\theaffil 19
thebibliography environment 75
\thecollab 19
\thempfn 711, 870
\thempfootnote 711, 870
\thispagestyle 818, 874
\tightenlines@sw 1090
\title 10, 13, 18, 55, 56
title, argument 10
\title 37, 200, 273, 1284
\title@column 828, 1189
\titleblock@produce . . 828, 866
titlepage (environment) . . . 816
\titlepage@sw . . 842, 1289, 1321
\toclevel@abstract 698
\toclevel@title 899
\today 10, 57
\true@sw 31, 32
\true@sw 479, 484, 1031
\tw@ 31, 45
\twocolumn@sw 817, 821
\twocolumngrid 821
\twoside@sw 822

U

\undefined 39
\unpenalty 1237
\unskip . . 603, 725, 905, 936, 958, 1223, 1237
\unsortedaddress 247
unsortedaddress document class option 11, 12, 16, 17, 47
\unvbox 805
\unvcopy 799
URL, argument 12
\url 26, 63, 69

V

\vadjust 739
\vbox 706
\volumenumber 1250
\volumeyear 1248
\vtop 799

X

\x@match 32
\x@match 519, 609

Y

\y@match 32
\y@match 618, 633
⟨*your document here*⟩ placeholder 9

Z

\z@ 49
\z@skip 739, 1038

Change History

4.0a

General: 4.0d had twoside option setting twoside switch to false . 3

comma not space between email and homepage 3

Initial version 3

Move after process options, so `\clearpage` not in scope of twocolumn 3

Move after process options, so the following test works 3

multiple preprint commands . . . 3

print homepage 3

protect against hyperref revtex kludges which are not needed now 3

Rearrange the ordering so numerical ones come first. AO: David, what does this mean? . . 3

single space footnotes 3

use font-dependent spacing 3

4.0b

`\@author@join`: Change handling of `\@author` to prevent parsing problems. (Helps in error recovery when syntax errors exist) 20

`\@pacs@produce`: Separate option now controls production 50

`\titleblock@produce`: Set `\footnote` to LaTeX standard version for this scope. 40

General: Added localization of `\figuresname` 3

Added localization of `\tablesname` 3

AO: all code for `10pt` is in this module. 3

AO: all code for `11pt` is in this module. 3

AO: all code for `12pt` is in this module. 3

AO: made aps.rtx part of revtex4.dtx 3

AO: remove duplicates 3

call `\print@floats` 3

Defer assignment until `\AtBeginDocument` time. 3

Defer decision until `\AtBeginDocument` time 3

Define three separate environments, defer assignment to `\AtBeginDocument` time. . . . 3

First modifications by Arthur Ogawa (mailto:arthur_ogawa at sbcglobal dot net) 3

Frank Mittelbach, has stated in `multicol`: “The kernel command `\@footnotetext` should not be modified.” Thus, I have removed David Carlisle’s redefinition of that command. Note, however, that later versions of `multicol` do not require this workaround. Belt and suspenders. 3

Gathered all code for the abstract environment together and abstracted out the formatting details for journals to override. 35

Move this “complex” option to the front, where it can be overridden by “simple” options. 3

New option 3

One-line caption sets flush left. . 3

only execute if appropriate 3

Processing delayed to `\AtBeginDocument` time 3

Removed invocation of nonexistent class option `groupauthors` and all other class options that should only be invoked by the document. (Otherwise precedence of class options does not work.) 3

Restore all media size class option of ␣classes.dtx 3

Stack `\preprint` args flush right at right margin. 3

4.0c

General: (AO, 115) If three or more preprints specified, set on

single line, with commas. 3
(AO, 129) section* within appendix was producing appendixname 3
*-form mandates pagebreak . . . 3
also spelled "acknowledgements". 3
Do not put by REVTeX in every page foot 3
grid changes via ltxgrid procedures 3
grid changes with ltxgrid 3
Insert procedure \checkindate . 3
Lose compatability mode. 3
New ltxgrid-based code, other bug fixes 3
New option "checkin" 3
Prevent an inner footnote from performing twice 3
titlepage: grid changes with ltxgrid 38
4.0d
\@affil@present@group: \comma@space 50
\@pacs@produce: PACS boxed to width of abstract 50
\altaffiliation: change name from \tempaffiliation 30
\maketitle: hookify 39
General: Also alter how lists get indented. 3
But alternative spelling is deprecated. 3
eprint takes an optional argument, syntactical only in this case. 3
make longtable trigger the head, too 3
More features and bug fixes: compatability with longtable and array packages. Now certainly incompatible with multicol. 3
New option 3
abstract: hookify 35
If empty \abstractname omit spacing 35
4.0e
\@keywords@produce: Keywords on same footing as PACS . . . 51
\@pacs@produce: Warn if no production 50
General: adornments above and below. 3
Bug fixes and minor new features: title block affiliations can have ancillary data, just like authors; clearpage processing revamped, with floats staying in order; widetext ornaments. 3
New option showkeys 3
4.0f
\@author@finish: Actually append \AU@grp to \@AAClist. 21
General: Last bug fixes before release. 3
4.0g
General: multiple preprint commands 35
4.0h
\@author@present@script: Changes to support groups of affiliations 45
\frontmatter@addressnewline: new hook 54
\move@AUAF: Changes to support groups of affiliations 26
4.0l
\altaffiliation: add optional argument handling back 30
\frontmatter@author@produce@script: Changes to add collaboration processing, which now means doing comma/and processing on each sublist 42
\set@listcomma@count: macro added 30
4.0rc1
General: grid changes with push and pop 3
Running headers always as if two-sided 3
4.0rc4
General: hyperref is no longer loaded via class option: use a usepackage statement instead . 3
4.1a
\affil@present@script: removed code that had been commented

out 44
\affils@present@script: removed code that had been commented out 44
\email: (AO, 496) remove default prependation to \email. 29
\homepage: (AO, 496) remove default prependation to \homepage. 30
General: (AO, 451) "Cannot have more than 256 cites in a document" 3
(AO, 457) Endnotes to be sorted in with numerical citations. . . . 3
(AO, 460) "Proper style is "FIG. 1. ..." (no colon)" 3
(AO, 461) Change the csname revtex uses from @dotsep to ltxu@dotsep. The former is understood in mu. (What we wanted was a dimension.) 3
(AO, 478) \ds@letterpaper, so that "letterpaper really is the default" 3
(AO, 488) Change processing of options to allow an unused option to specify society and journal 3
For natbib versions before 8.21, \NAT@sort was consulted only as natbib was being read in. Now it is fully dynamic. 3
abstract: (AO, 443) create an entry in the PDF bookmarks for the abstract via the toc . . 35
4.1b
\@affil@join: Use \@affil@def for all access to \@affil 23
\@affil@match: (AO, 486) \collaboration to work with groupedaddress 25
\@author@finish: (AO, 486) \collaboration to work with groupedaddress 21
Provide for diagnostics via \frontmatterverbose@sw . . . 21
Use \@affil@def for all access to \@affil 21
\@author@join: (AO, 486) \collaboration to work with groupedaddress 20
Use \frontmatter@init as the vehicle for all initialization . . 20
\@published: Make the syntax of this package switch optional . 34
\@shorttitle: Use \frontmatter@init as the vehicle for all initialization . . 18
\add@AUCO@grp: (AO, 486) \collaboration to work with groupedaddress 20
Provide for diagnostics via \frontmatterverbose@sw . . . 21
\affiliation: Use \@affil@def for all access to \@affil 22
\affils@present@group: Provide for diagnostics via \frontmatterverbose@sw . . . 47
\author: Make the syntax of this package switch optional 19
\c@collab: (AO, 486) \collaboration to work with groupedaddress 19
\collaboration: (AO, 486) \collaboration to work with groupedaddress 19
\do@title: Make the syntax of this package switch optional . 18
\doauthor: Provide facility \affil@cutoff: if there is only one affiliation on the title page, no superscript 31
\frontmatter@author@produce@group: Provide for diagnostics via \frontmatterverbose@sw . . . 47
\frontmatter@author@produce@script: Provide for diagnostics via \frontmatterverbose@sw . . . 41
\frontmatter@title@produce: (AO, 443) PDF Bookmark for title page 41
\move@AF: (AO, 486) \collaboration to work with groupedaddress 24
Provide for diagnostics via \frontmatterverbose@sw . . . 25
\move@AUAF: (AO, 486) \collaboration to work with groupedaddress 26

Change csname \AFS@opr to \AFG@opr 26
Provide for diagnostics via \frontmatterverbose@sw . . . 26
\noaffiliation: Use \@affil@def for all access to \@affil 23
\thanks: Make the syntax of this package switch optional 30
General: (AO) Implement bibnotes through \frontmatter@footnote@produce instead of \bibnotes@sw 3
(AO) No longer need to test \chapter as of natbib version 8.2 3
(AO) No longer use \secnumarabic@sw, instead use \setup@secnums 3
(AO) Provide more diagnostics when \@society is assigned. . . 3
(AO) Structure the Abstract using the bibliography environment 3
(AO) coordinate \if@twoside with \twoside@sw 3
(AO) make settings at class time instead of deferring them to later. 3
(AO) provide option longbibliography 3
(AO, 455) Be nice to a list within the abstract (assign \@totalleftmargin). 3
Add \@hangfroms@section 3
Add option reprint, opposite of preprint, and preferred alternative to twocolumn 3
As with author formatting, rag the right more, and assign \@totalleftmargin. Also neutralize \def@after@address. 3
Break out \@caption@fignum@sep 3
Class option galley sets \preprintsty@sw to false 3
Code relating to new syntax for frontmatter has been placed in ␣ ltxfront.dtx 3
Make the syntax of this package switch optional 55
Package textcase is now simply a required package 3
Procedures \@parse@class@options@society and \@parse@class@options@journal and friends 3
Rag the right even more: .8\hsize. Also, assign \@totalleftmargin. 3
Read in all required packages together 3
Remove options newabstract and oldabstract 3
Section numbering via procedures \secnums@rtx and \secnums@arabic. 3
The rmp journal substyle selects groupedaddress by default. . . . 3
The csname substyle@ext is now defined without a dot (.), to be compatible with LaTeXusage (see @clsextension and @pkgextension). 3
Use \setup@hook to initialize all. 3
frontmatter@description: (AO, 455) provide a description environment for the abstract . 37
titlepage: Make the syntax of this package switch optional . 38
4.1c
\def@after@address@empty: Parametrize the production after the address 49
\maketitle@Hy: Hyperref compatibility: take \maketitle back. 39
\titleblock@produce: Produce PACS, keywords only if specified 40
General: Document class option longbibliography via \substyle@post 3
4.1d
\@affil@group: (AO, 410) compare \@tempa to \blankaffiliation. 50

General: Definition of \@fnsymbol follows fixltx2e.sty 3
4.1e
General: (AO, 455) be nice to a list within the abstract 3
4.1f
\frontmatter@footnotetext: (AO, 515) Use \set@footnotefont, which is defined in ltxutil 36
General: (AO, 513) Add class option linenumbers: number the lines a la lineno 3
(AO, 516) Merged references are separated with a semicolon . . . 3
(AO, 520) Automatically produce \bibliography command when needed 3
(AO, 521) Lonely bibliography head 3
(AO, 522) Warn if software is expired 3
(AO, 523) Add class option nomerge, to turn off new natbib 8.3 syntax 3
(AO, 524) Makes no sense if citations are superscript numbers and so are footnotes . 3
(AO, 530) \@fnsymbol: Failed to import fixltx2e.sty technology. Return to LaTeX core. 3
4.1g
\@author@finish: (AO, 535) Robustify \footnote while \class@warn is expanding its argument 21
General: (AO, 525) Remove phantom paragraph above display math that is given in vertical mode 3
(AO, 538) \MakeTextUppercase inappropriately expands the double backslash 3
(AO, 539) Robustify \class@warn against TOC processing 55
4.1h
General: (AO) Remove expiry code in the release software 3
4.1i
\@author@finish: (AO, 540) More runtime diagnostics via \frontmatterverbose@sw . . . 21
\@author@join: (AO, 540) More runtime diagnostics via \frontmatterverbose@sw . . . 20
\affiliation: (AO, 540) More runtime diagnostics via \frontmatterverbose@sw . . . 22
\frontmatter@author@produce@group: (AO, 540) More runtime diagnostics via \frontmatterverbose@sw . . . 47
\frontmatter@author@produce@script: (AO, 540) More runtime diagnostics via \frontmatterverbose@sw . . . 42
\move@AF: (AO, 540) Rationalize code that appends to \AF@grp 24, 25
\move@AUAF: (AO, 540) More runtime diagnostics via \frontmatterverbose@sw . . . 26
\noaffiliation: (AO, 540) More runtime diagnostics via \frontmatterverbose@sw . . . 23
General: (AO, 541) Defer assignment of \cite until after natbib loads 3
4.1j
General: (AO, 545) hypertext capabilities off by default; enable with hypertext 3
(AO, 549) Repairing natbib's \BibitemShut and \bibAnnote 3
(AO, 552) Repair spacing in \onlinecite 3
4.1k
General: (AO, 554) give the \newlabel command syntax appropriate to the hyperref package 3
4.1m
\@author@finish: (AO, 563) \collaboration to work with both groupedaddress and superscriptaddress 21
\@collaboration@gobble: (AO, 563) Let

\@collaboration@gobble
parse all three arguments of \CO@opr 31
\frontmatter@author@produce@group: (AO, 563) More complete runtime diagnostics via \frontmatterverbose@sw . . . 47
\frontmatter@author@produce@script: (AO, 563) More complete runtime diagnostics via \frontmatterverbose@sw . . . 42
\move@AUAF: (AO, 563) \collaboration to work with both groupedaddress and superscriptaddress 27
4.1n
\@affil@join: (AO, 567) Handle \noaffiliation case 23
\@affil@present@script@: (AO, 567) Handle punctuation for superscriptaddress style authors' extra information, such as \email 46
\@author@finish: (AO, 567) Handle \noaffiliation case . 21
\@author@present@script: (AO, 567) Handle \noaffiliation case 45
\doauthor: (AO, 567) Handle punctuation for superscriptaddress style authors' extra information, such as \email 31
\frontmatter@footnote: (AO, 572) Independent footnote counter for title block. Abstract footnote counter shared with body. 33
\frontmatter@footnote@produce@endnote: (AO, 572) Independent footnote counter for title block. Abstract footnote counter shared with body. 51
\frontmatter@footnotetext: (AO, 571) allow split after last line of footnote 36
(AO, 572) Independent footnote counter for title block. Abstract footnote counter shared with body. 36
\frontmatter@makefntext: (AO, 572) Independent footnote counter for title block. Abstract footnote counter shared with body. 52
\frontmatter@thefootnote: (AO, 572) Independent footnote counter for title block. Abstract footnote counter shared with body. 52
\homepage: (AO) remove http:// from \href call 30
\move@AF: (AO, 567) Handle \noaffiliation case 24
\noaffiliation: (AO, 567) Handle \noaffiliation case . 23
\titleblock@produce: (AO, 572) Independent footnote counter for title block. Abstract footnote counter shared with body. 40
General: (AO) fine-tune spacing above and below widetext 3
(AO, 565) restore 4.0 behavior: invoking class option preprint implies class option preprintnumbers 3
(AO, 566) restore 4.0 behavior: flush column bottoms 3
(AO, 569) Use of hyperref interferes with column balancing of last page 3
(AO, 569) execute the after-last-shipout procedures from within the safety of the output routine 3
(AO, 571) Interface \set@footnotewidth for determining the set width of footnotes 3
(AO, 571) class file must set \splittopskip; fine tune \skip\footins; \footnoterule defined in terms of \skip\footins 3
(AO, 572) Independent footnote counter for title block. Abstract footnote counter shared with body. 3, 56, 57

(AO, 572) \@makefntext and \frontmatter@makefntext must be defined harmoniously . 3
(AO, 573) arrange to load lineno after any other packages. 3
(AO, 575) the default for journal prstper is longbibliography . . . 3
(AO, 576) In .bst files, remove support for the annote field . . . 3
4.1o
General: (AO, 549) Remove patch to natbib, which is now at version 8.31a 3
(AO, 575) Automatically incorporate the (BibTEX-generated) .bbl into an explicit thebibliography . . 3
(AO, 578) accommodate the possible space character preceding \BibitemShut. 3
(AO, 579) Endnote shall comprise their own BibTEX entry type: @FOOTNOTE. . . . 3
(AO, 580) Control .bst at run time. 3
(AO, 580) Provide a document class option to turn off production of eprint field in bibliography. 3
(AO, 581) Handle case: merged references, with first ending in a stop character. 3
4.1p
General: (AO, 583) Provide interface to ltxgrid \onecolumn@grid@setup and \twocolumn@grid@setup 3
(AO, 584) Per MD, remove trailing space character from each journal abbreviation: it had caused an extraneous space in the .bbl 3
4.1q
General: (AO, 586) When .bbl is pasted into the document, prevent automatic bibliography inclusion. 3
(AO, 588) Only write REVTEX-specific BibTeX .bib data if the .bst style is set by REVTeX. 3
4.1r
General: (AO, 595) Provide \lovname along with other List of Videos definitions. 3